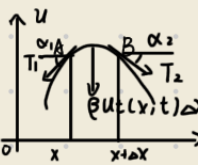


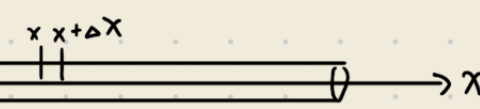
习题1

1.  竖直方向:  $\rho \Delta x U_{tt} = -T_1 \sin \alpha_1 - T_2 \sin \alpha_2 - b U_t(x, t) \Delta x$   
代入  $\Delta \sin \alpha_1, \sin \alpha_1 \sim \tan \alpha_1 \sim U_x(x, t), \sin \alpha_2 \sim \tan \alpha_2 \sim -U_x(x + \Delta x, t), T_1 \sim T_2 \sim T_0$   
 $\therefore \rho \Delta x U_{tt} = T_0 [U_x(x + \Delta x, t) - U_x(x, t)] - b U_t(x, t) \Delta x$   
 $U_x(x + \Delta x, t) - U_x(x, t) = U_{xx}(x, t) \Delta x$   
 $\Rightarrow \rho U_{tt} = T_0 U_{xx} - b U_t \Rightarrow U_{tt} + \frac{b}{\rho} U_t = a^2 U_{xx}$

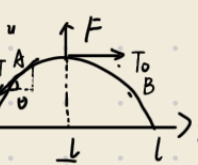
2. 左端温度为0 右端温度为0  
$$\begin{cases} U_t = a^2 U_{xx}, 0 < x < l, t > 0 \\ U(0, t) = 0, k U_x(l, t) = q, t \geq 0 \\ U(x, 0) = \varphi(x), 0 \leq x \leq l \end{cases}$$
$$\begin{cases} U_t = a^2 U_{xx}, 0 < x < l, t > 0 \\ -k U_x(0, t) = q, U(l, t) = 0, t \geq 0 \\ U(x, 0) = \varphi(x), 0 \leq x \leq l \end{cases}$$

3. 
$$\begin{cases} U_t = a^2 U_{xx}, (x, y, z) \in B, t > 0 \\ -k U = h(U - 27), (x, y, z) \in \partial B, t \geq 0 \\ U(x, y, z, 0) = 20, (x, y, z) \in \bar{B} \end{cases}$$

4. 左端封闭 右端封闭  
$$\begin{cases} U_t = a^2 U_{xx}, 0 < x < l, t > 0 \\ U(0, t) = 0, k U_x(l, t) = h(U_0 - U(l, t)), t \geq 0 \\ U(x, 0) = 0, 0 \leq x \leq l \end{cases}$$
$$\begin{cases} U_t = a^2 U_{xx}, 0 < x < l, t > 0 \\ -k U_x(0, t) = h(U_0 - U(0, t)), t \geq 0 \\ U(x, 0) = 0, 0 \leq x \leq l \end{cases}$$

5.   
 $m \Delta x = \frac{\partial(\rho(x, t) \Delta x)}{\partial t} = \Delta x \rho_t$   
单位时间内通过x处流入:  $\rho(x, t) V(x, t) \Delta x$   
单位时间内通过x+Δx处流出:  $\rho(x + \Delta x, t) V(x + \Delta x, t) \Delta x$   
 $\therefore$  单位时间内净增量  $\rho(x, t) V(x, t) \Delta x - \rho(x + \Delta x, t) V(x + \Delta x, t) \Delta x$   
微分中值定理:  $\rho V(x + \Delta x, t) - \rho V(x, t) = (\rho V)_x \Delta x$   
质量守恒原理:  $\Delta x \rho_t = -\Delta x (\rho V)_x$  即  $\rho_t + (\rho V)_x = 0$

6. (1) 
$$\begin{cases} U_t = a^2 U_{xx}, 0 < x < l, t > 0 \\ U(0, t) = U_0, k U_x(l, t) = \sin \omega t, t \geq 0 \\ U(x, 0) = 0, 0 \leq x \leq l \end{cases}$$
  
(2) 
$$\begin{cases} U_t = a^2 U_{xx} + \frac{x}{t}, 0 < x < l, t > 0 \\ U(0, t) = U_0, k U_x(l, t) = \sin \omega t, t \geq 0 \\ U(x, 0) = 0, 0 \leq x \leq l \end{cases}$$

7. 
$$\begin{cases} U_{tt} = a^2 U_{xx}, 0 < x < l, t > 0 \\ U(0, t) = 0, U(l, t) = 0, t \geq 0 \\ U(x, 0) = \varphi(x), U_t(x, 0) = 0, 0 \leq x \leq l \end{cases}$$
  
对于  $\varphi(x)$  的求解  
  
对于A点, 水平方向:  $T \cos \theta = T_0$ , 竖直方向:  $T \sin \theta = F$   
 $\Rightarrow \tan \theta = \frac{F}{T_0} = U_x(x, 0) = \frac{d\varphi}{dx} \Rightarrow \varphi(x) = \frac{F}{T_0} x, 0 \leq x < \frac{l}{2}$   
对于B点, 水平方向:  $\frac{F}{T_0} = -U_x(x, 0) = -\frac{d\varphi}{dx} \Rightarrow \varphi(x) = \frac{F}{T_0} (l - x), \frac{l}{2} \leq x \leq l$

综上: 
$$\begin{cases} U_{tt} = a^2 U_{xx}, 0 < x < l, t > 0 \\ U(0, t) = 0, U(l, t) = 0, t \geq 0 \\ U(x, 0) = \begin{cases} \frac{F}{T_0} x, 0 \leq x < \frac{l}{2} \\ \frac{F}{T_0} (l - x), \frac{l}{2} \leq x \leq l \end{cases}, U_t(x, 0) = 0, 0 \leq x \leq l \end{cases}$$

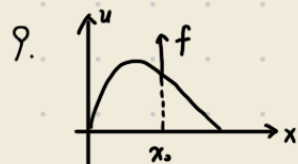


均取向上为正

$$T_0 u_x(l, t) = f(t) - k(u(l, t) - L)$$

$$\Rightarrow T_0 u_x + k u = f(t) + k L, t \geq 0$$

当  $f(t) = 0$ , 即不加外力时  $T_0 u_x + k u = k L, t \geq 0$



设施加  $f$  的点的横坐标为  $x_0$

取  $x \rightarrow x + \Delta x$  的一小段弧  $\Delta x \rightarrow 0$

$$\rho \Delta x u_{tt} = \rho \Delta x a^2 u_{xx} + f \Rightarrow u_{tt} = a^2 u_{xx} + g$$

$$\text{其中 } g = \lim_{\Delta x \rightarrow 0} \frac{f(t)}{\rho \Delta x} = \frac{f}{\rho} \delta(x - x_0)$$

$$\therefore \begin{cases} u_{tt} = a^2 u_{xx} + \frac{f}{\rho} \delta(x - x_0), 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), 0 \leq x \leq l \end{cases}$$

$$10. u_x = 2x, u_y = 2y, y u_x - x u_y = 0$$

$\therefore u = x^2 + y^2$  是该方程的解

$$\text{对于 } u = \varphi(x^2 + y^2), u_x = 2x\varphi', u_y = 2y\varphi'$$

$$y u_x - x u_y = 0 \therefore u = \varphi(x^2 + y^2) \text{ 是该方程的解}$$

$$11. u_x = 1, u_y = \varphi'(y), u_{xy} = 0, u_{xx} = 0 \quad u_x u_{xy} - u_y u_{xx} = 0$$

$$\therefore u = x + \varphi(y) \text{ 是方程 } \frac{\partial u}{\partial x} \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial u}{\partial y} \frac{\partial^2 u}{\partial x^2} = 0 \text{ 的一个解}$$

$$u = \varphi(x + \varphi(y))$$

$$u_x = \varphi', u_y = \varphi' \varphi'(y), u_{xy} = \varphi'(y) \varphi'', u_{xx} = \varphi''$$

$$u_x u_{xy} - u_y u_{xx} = 0 \therefore u = \varphi(x + \varphi(y)) \text{ 仍是原方程的解}$$

该方程是二次非齐方非线性方程

$$12. u(x, y) = x^2 y + y^2 + \varphi(x)$$

$$u(x, x^2) = x^4 + x^4 + \varphi(x) = 1 \Rightarrow \varphi(x) = 1 - 2x^4$$

$$\therefore u(x, y) = x^2 y + y^2 - 2x^4 + 1$$

13.

$$(1) u_{1t} = -a f', u_{1x} = f', u_{1tt} = a^2 f'', u_{1xx} = f'', u_{1tt} = a^2 u_{1xx}$$

$$u_{2t} = a g', u_{2x} = g', u_{2tt} = a^2 g'', u_{2xx} = g'', u_{2tt} = a^2 u_{2xx}$$

$u_1(x, t) = f(x - at), u_2(x, t) = g(x + at)$  都是一维波动方程  $u_{tt} = a^2 u_{xx}$  的解

$$(2) \text{ 设 } u(x, t) = f(x - at) + g(x + at), u_t = -a f'(x - at) + a g'(x + at)$$

$$\begin{cases} u(x, 0) = f(x) + g(x) = \sin x \\ u_t(x, 0) = -a f'(x) + a g'(x) = 0 \end{cases} \Rightarrow f(x) = g(x) = \frac{\sin x}{2}$$

$$\therefore u(x, t) = \frac{1}{2} [\sin(x - at) + \sin(x + at)] = \sin x \cos at$$

14.

$$(1) w(x, t) = \frac{\sin t - t}{2} x + t$$

$$(2) w(x, t) = (1 + t^2) x + 1$$

$$(3) w(x, t) = \frac{\psi(t) - \varphi(t)}{2 \times 3} x^2 + \varphi(t) x = \frac{\psi(t) - \varphi(t)}{6} x^2 + \varphi(t) x$$

(4) 设为抛物线形式  $w(x, t) = c_1 x^2 + c_2 x + c_3$ , 令  $w(2, t) = 0$

$$w_x = 2C_1 x + C_2$$

$$\begin{cases} w(2, t) = 4C_1 + 2C_2 + C_3 = 0 \\ w_x(2, t) = C_2 = t^2 \\ w_x(2, t) + w(2, t) = 4C_1 + C_2 = t \end{cases}$$

$$\Rightarrow \begin{cases} C_1 = \frac{t-t^2}{4} \\ C_2 = t^2 \\ C_3 = -(t^2+t) \end{cases} \Rightarrow w(x, t) = \frac{t-t^2}{4} x^2 + t^2 x - (t^2+t)$$

15.

$$(1) u_t - a^2 u_{xx} = xt \Rightarrow w_t - a^2 w_{xx} = xt$$

$$\text{令 } w_{xx} = 0 \Rightarrow w(x, t) = \frac{1}{2} x t^2$$

$$(2) w_{xx} + w_{yy} = \sin x + xy \Rightarrow w(x, y) = -\sin x + \frac{1}{6} xy^3$$

22. 设铁杆温度是  $u(x, t)$ , 铜杆温度是  $v(x, t)$

$$\text{边界条件 } -k u_x(0, t) = q, v(2l, t) = u_0, t \geq 0$$

研究  $x=l$  处: 取邻域  $x_1 < l < x_2$

$$\text{在 } [x_1, x_2] \text{ 区间中, 热量守恒: } \rho_1 C_1 (l - x_1) + \rho_2 C_2 (x_2 - l) = -S_0 [k_1 u_x(x_1, t) - k_2 v_x(x_2, t)]$$

$$\text{取极限 } x_2 \rightarrow l^+, x_1 \rightarrow l^- \Rightarrow k_1 u_x(l, t) = k_2 v_x(l, t)$$

$$x=l \text{ 处温度相同 } u(l, t) = v(l, t)$$

$$\text{综上: } \begin{cases} u_t = a_1^2 u_{xx}, 0 < x < l, t > 0; v_t = a_2^2 v_{xx}, l < x < 2l, t > 0 \\ -k u_x(0, t) = q, v(2l, t) = u_0, t \geq 0 \\ u(x, 0) = \varphi(x), 0 \leq x \leq l, v(x, 0) = \varphi(x), l \leq x \leq 2l \\ k_1 u_x(l, t) = k_2 v_x(l, t), u(l, t) = v(l, t), t > 0 \end{cases}$$

$$23. \begin{cases} u_t = a^2 u_{xx}, 0 < x < \infty, t > 0 \\ u(at, t) = g(t), t \geq 0 \\ u(x, 0) = \varphi(x), 0 \leq x < \infty \end{cases}$$

24. 取小时间段  $\Delta t$  和小区间  $\Delta x$

设  $t$  时刻金属已燃烧的长度为  $s(t)$  则燃烧速度为  $s'(t)$

$\Delta t$  时间内燃烧需要的热量:  $L \rho s'(t) \Delta t$

$$\text{热量守恒: } L \rho s'(t) \Delta t = \int_t^{t+\Delta t} k u_{xx}(s(t) + \Delta x, t) dt + q \Delta t$$

$$\text{令 } \Delta t \rightarrow 0, \Delta x \rightarrow 0 \Rightarrow L \rho s'(t) = k u_{xx}(s(t), t) + q$$

$$\begin{cases} u_t = a^2 u_{xx}, s(t) < x < l, t > 0 \\ L \rho s'(t) = k u_x(s(t), t) + q, u(s(t), t) = A, t \geq 0 \\ u_x(l, t) = 0, t \geq 0 \\ u(x, 0) = \varphi(x), 0 \leq x \leq l \end{cases}$$

25. 设水部分温度为  $u(x, t)$ , 冰部分温度为  $v(x, t)$ , 水和冰的线密度均为 1

取包含冰水界面的小区间  $[p(t), p(t) + \Delta x]$

$$-L \rho' \Delta t = -\int_t^{t+\Delta t} k_2 v_x(p(t), t) dt + \int_t^{t+\Delta t} k_1 u_x(p(t) + \Delta x, t) dt$$

$$\Delta t \rightarrow 0, \Delta x \rightarrow 0: L \rho' \Delta t = -k_1 u_x(p(t), t) + k_2 v_x(p(t), t)$$

此即冰水界面  $x=p(t)$  处的边界条件

$$\therefore \text{定解问题} \begin{cases} u_t = a_1^2 u_{xx}, & 0 < x < \infty, \quad t > 0 \\ v(0, t) = G, & t \geq 0 \\ v(\rho(t), t) = u(\rho(t), t) = 0, & t > 0 \\ L\rho'(t) = -k_1 u_x(\rho(t), t) + k_2 v_x(\rho(t), t), & t > 0 \\ u(x, 0) = C, & 0 \leq x < \infty \end{cases}$$

26. 推导左侧的边界条件, 取小区间  $[0, \Delta x]$

$$\rho \Delta x \frac{u}{t} \Big|_t^{t+\Delta t} = k u_x(\Delta x, t + \theta \Delta t) \Delta t - \sigma [u^4(0, t + \theta \Delta t) - 10^4] \Delta t, \quad 0 < \theta < 1$$

$$\text{令 } \Delta t \rightarrow 0, \Delta x \rightarrow 0 \Rightarrow 0 = k u_x(0, t) - \sigma u^4(0, t) + 10^4 \sigma$$

$$\text{即 } \sigma [10^4 - u^4(0, t)] = k u_x(0, t)$$

$$\therefore \text{定解问题} \begin{cases} u_t = a^2 u_{xx}, & 0 < x < l, \quad t > 0 \\ \sigma [10^4 - u^4(0, t)] = k u_x(0, t), & u_x(l, t) = 0, \quad t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

28.

(1) 对于一个小弧段的动能:  $dE_k = \frac{1}{2} \rho u_t^2 dx$

$$E_k(t) = \int dE_k = \frac{\rho}{2} \int_0^l u_t^2 dx$$

(2) 对于一小段弧, 在  $dt$  时间内张力  $T$  所做功为  $dW = dE_k = \frac{1}{2} \rho u_t^2 dx$

$$u_t^2 = u_x^2 \cdot \left(\frac{dx}{dt}\right)^2 = u_x^2 \cdot \frac{T}{\rho} \therefore E_p(t) = \int dW = \frac{T}{2} \int_0^l u_x^2 dx$$

$$(3) E(t) = E_k(t) + E_p(t) = \frac{\rho}{2} \int_0^l u_t^2 dx + \frac{T}{2} \int_0^l u_x^2 dx$$

29.

$$(1) E_k(t) = \frac{\rho}{2} \int_0^l u_t^2 dx$$

$$(2) \text{弦线因弹性形变产生的势能: } E_p(t) = \frac{T}{2} \int_0^l u_x^2 dx$$

弹性支撑边界产生的势能:

$x=0$  处相当于  $k=1$  的弹簧,  $x=l$  处相当于  $k=2$  的弹簧

$$E_{p2}(t) = \frac{1}{2} u^2(0, t) + \frac{1}{2} \times 2 u^2(l, t) = \frac{1}{2} u^2(0, t) + u^2(l, t)$$

$$\therefore E_p = E_{p1} + E_{p2} = \frac{T}{2} \int_0^l u_x^2 dx + \frac{1}{2} u^2(0, t) + u^2(l, t)$$

补充题

将如下定解问题的非齐次边界齐次化

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, \quad t > 0 \\ u(0, t) = u_0, & u_x(l, t) = \sin \omega t, \quad t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

$$\text{令 } w(x, t) = \sin \omega t x + u_0$$

$$w_x(x, t) = \sin \omega t, \quad w_{xx} = 0$$

$$w_t(x, t) = \omega \cos \omega t x$$

$$u_t - a^2 u_{xx} = 0 \Rightarrow (v_t + w_t) - a^2 (v_{xx} + w_{xx}) = 0 \Rightarrow v_t - a^2 v_{xx} = -\omega x \cos \omega t, \quad 0 < x < l, \quad t > 0$$

$$v(0, t) = 0, \quad v(l, t) = 0, \quad t \geq 0$$

$$v(x, 0) + w(x, 0) = u_0 \Rightarrow v(x, 0) = 0, \quad 0 \leq x \leq l$$

$\therefore$  齐次化边界后的定解问题

$$\begin{cases} v_t - a^2 v_{xx} = -\omega x \cos \omega t, & 0 < x < l, \quad t > 0 \\ v(0, t) = 0, & v(l, t) = 0, \quad t \geq 0 \\ v(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

习题 2

1. 令  $u(x, t) = X(x)T(t)$ , 得特征值问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, 0 < x < l \\ X(0) = 0, X'(l) = 0 \end{cases} \Rightarrow \lambda_n = \left(\frac{2n+1}{2l}\pi\right)^2, X(x) = \sin \frac{2n+1}{2l}\pi x, n = 0, 1, 2, 3, \dots$$

$$\varphi(x) = \sum_{n=0}^{\infty} \varphi_n X_n(x), \varphi_n = \frac{2}{l} \int_0^l \varphi(\alpha) \sin \frac{2n+1}{2l}\pi \alpha d\alpha$$

$$\psi(x) = \sum_{n=0}^{\infty} \psi_n X_n(x), \psi_n = \frac{2}{l} \int_0^l \psi(\alpha) \sin \frac{2n+1}{2l}\pi \alpha d\alpha$$

$$f(x, t) = \sum_{n=0}^{\infty} f_n(t) X_n(x), f_n(t) = \frac{2}{l} \int_0^l f(\alpha, t) \sin \frac{2n+1}{2l}\pi \alpha d\alpha$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x), \text{代入原方程}$$

$$\sum_{n=0}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) X_n''(x) = \sum_{n=0}^{\infty} f_n(t) X_n(x), X_n'' = -\lambda_n X_n(x)$$

$$\Rightarrow T_n''(t) + a^2 \lambda_n T_n(t) = f_n(t), n = 0, 1, 2, \dots$$

$$\sum_{n=0}^{\infty} T_n(0) X_n(x) = \sum_{n=0}^{\infty} \varphi_n X_n(x) \Rightarrow \varphi_n = T_n(0), n = 0, 1, 2, \dots$$

$$\sum_{n=0}^{\infty} T_n'(0) X_n(x) = \sum_{n=0}^{\infty} \psi_n X_n(x) \Rightarrow \psi_n = T_n'(0), n = 0, 1, 2, \dots$$

求解关于  $T_n(t)$  的定解问题:

$$T_n(t) = \varphi_n \cos \frac{n\pi a}{l} t + \frac{\psi_n \cdot l}{n\pi a} \sin \frac{n\pi a}{l} t + \frac{l}{n\pi a} \int_0^t f_n(\tau) \sin \frac{n\pi a}{l}(t-\tau) d\tau, n = 0, 1, 2, \dots$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} T_n(t) \cdot \sin \frac{2n+1}{2l}\pi x, T_n(t) \text{ 由上式给出}$$

2.

(1),  $\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}\right)x, n = 0, 1, 2, \dots$

(2), 令  $x+1 = t \Rightarrow \begin{cases} X''(t) + \lambda X(t) = 0, 0 < t < 2 \\ X(0) = 0, X(2) = 0 \end{cases}$

$$\Rightarrow \lambda_n = \left(\frac{n\pi}{2}\right)^2, X_n(x) = \sin \frac{n\pi}{2}(x+1), n = 0, 1, 2, \dots$$

(3)  $\lambda_n = \left(\frac{2n+1}{2l}\pi\right)^2, X_n(x) = \cos \frac{2n+1}{2l}\pi x, n = 0, 1, 2, \dots$

(4) ①  $\lambda = 0$  时,  $X_0(x) = c_1 + c_2 x, c_2 = 0, X(x) = c_1 \neq 0 \therefore \lambda = 0$  是特征值

②  $\lambda \neq 0$  时  $X(x) = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x, X'(x) = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$

$$\begin{cases} X(0) = X(2l) \\ X'(0) = X'(2l) \end{cases} \Rightarrow \lambda = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \left\{ \cos \frac{n\pi}{l} x, \sin \frac{n\pi}{l} x \right\} n = 1, 2, 3, \dots$$

综上:  $\lambda_n = \left(\frac{n\pi}{l}\right)^2, n = 0, 1, 2, \dots$

$$X_0(x) = 1, X_n(x) = \left\{ \cos \frac{n\pi}{l} x, \sin \frac{n\pi}{l} x \right\}, n = 1, 2, 3, \dots$$

3.

$$(1) X''(x) + \lambda X(x) = 0 \Rightarrow \int_0^l X''(x) X(x) dx + \lambda \int_0^l X'(x) dx = 0$$

$$\Rightarrow X'(x) X(x) \Big|_0^l - \int_0^l [X'(x)]^2 dx + \lambda \int_0^l X^2(x) dx = 0$$

$$\because X(0) = 0, X'(l) + X(l) = 0 \Rightarrow -X^2(l) - \int_0^l [X'(x)]^2 dx + \lambda \int_0^l X^2(x) dx = 0$$

$$\Rightarrow \lambda = \frac{X^2(l) + \int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) dx} \geq 0$$

若  $\lambda = 0$ , 则  $X(x) = c_1 + c_2 x, X(0) = c_1 = 0, X'(x) = c_2$

$$X'(l) + X(l) = c_2(l+1) = 0 \Rightarrow c_2 = 0, \therefore \lambda = 0 \text{ 不是特征值}$$

$$\therefore \lambda > 0$$

(2) 设  $\lambda_1, \lambda_2$  对应的特征函数分别为  $X_1(x), X_2(x)$

$$\begin{cases} X_1''(x) + \lambda_1 X_1(x) = 0 \\ X_2''(x) + \lambda_2 X_2(x) = 0 \end{cases} \Rightarrow X_1''(x)X_2(x) - X_2''(x)X_1(x) + (\lambda_1 - \lambda_2)X_1(x)X_2(x) = 0$$

$$\Rightarrow \int_0^l X_1(x)X_2(x)dx = \frac{1}{\lambda_1 - \lambda_2} \int_0^l [X_2''(x)X_1(x) - X_1''(x)X_2(x)]dx$$

$$= \frac{1}{\lambda_1 - \lambda_2} [(X_2'X_1(x) - X_1'X_2(x)) \Big|_0^l - \int_0^l X_1'(x)X_2'(x) - X_1'(x)X_2'(x)dx] = 0$$

$\therefore$  不同的特征值对应的特征函数是正交的

(3)  $X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x, X'(x) = \sqrt{\lambda} (-C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x)$

$$\begin{cases} X(0) = C_1 = 0 \\ X'(l) + X(l) = C_2 \sin \sqrt{\lambda} l + \sqrt{\lambda} C_2 \cos \sqrt{\lambda} l = 0 \end{cases} \Rightarrow \tan \sqrt{\lambda} l = -\sqrt{\lambda}$$

方程  $\tan x = -\frac{x}{l}$  在区间上有唯一正根, 记为  $\mu_n, n=1, 2, 3, \dots$

$$\therefore \lambda_n = \left(\frac{\mu_n}{l}\right)^2, X_n(x) = \sin \frac{\mu_n}{l} x, n=1, 2, 3, \dots$$

4. (1)  $-\frac{d}{dx} [p(x) \frac{d}{dx} X(x)] + q(x)X(x) = \lambda X(x)$

$$\Rightarrow -\frac{d}{dx} [p(x) X'(x)] X(x) + q(x)X(x) = \lambda X^2(x)$$

$$\Rightarrow \int_0^l -\frac{d}{dx} [p(x) X'(x)] X(x) + q(x)X(x) dx = \lambda \int_0^l X^2(x) dx$$

$$\therefore X(0) = 0, X(l) = 0$$

$$\therefore \int_0^l -\frac{d}{dx} [p(x) X'(x)] X(x) + q(x)X(x) dx$$

$$= -[p(x) X'(x) X(x)] \Big|_0^l - \int_0^l p(x) X'^2(x) dx + \int_0^l q(x)X(x) dx$$

$$= \int_0^l p(x) X'^2(x) dx + q(x)X(x) dx$$

$$\therefore \lambda = \frac{\int_0^l p(x) X'^2(x) dx + q(x)X(x) dx}{\int_0^l X^2(x) dx} \geq 0$$

(2) 设  $\lambda_1, \lambda_2$  对应的特征函数分别为  $X_1(x), X_2(x)$

$$\begin{cases} -\frac{d}{dx} [p(x) \frac{d}{dx} X_1(x)] + q(x)X_1(x) = \lambda_1 X_1(x) \\ -\frac{d}{dx} [p(x) \frac{d}{dx} X_2(x)] + q(x)X_2(x) = \lambda_2 X_2(x) \end{cases}$$

$$\Rightarrow -\frac{d}{dx} [p(x) X_1'(x)] X_2(x) + \frac{d}{dx} [p(x) X_2'(x)] X_1(x) = (\lambda_1 - \lambda_2) X_1(x) X_2(x)$$

$$\therefore \int_0^l X_1(x)X_2(x)dx = \frac{1}{\lambda_1 - \lambda_2} \int_0^l \left[ -\frac{d}{dx} [p(x) X_1'(x)] X_2(x) + \frac{d}{dx} [p(x) X_2'(x)] X_1(x) \right] dx$$

$$= \frac{1}{\lambda_1 - \lambda_2} [-p(x) X_1'(x) X_2(x) + p(x) X_2'(x) X_1(x)] \Big|_0^l = 0$$

$\therefore$  不同的特征值对应的特征函数是正交的

5.

(1) 设  $u(x, t) = X(x)T(t)$

$$\Rightarrow X(x)T''(t) - a^2 X''(x)T(t) = 0 \Rightarrow \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$\Rightarrow \begin{cases} T''(t) + a^2 \lambda T(t) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

齐次边界  $T(t)$  不恒为 0  $\Rightarrow X'(0) = X'(l) = 0$

特征值:  $\lambda_n = (\frac{n\pi}{l})^2$ , 特征函数:  $X_n(x) = \cos \frac{n\pi}{l} x$ ,  $n=0, 1, 2, \dots$

分离变量的齐次解:  $u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n \cos \frac{n\pi}{l} x$

初值常微分方程:  $\sum_{n=0}^{\infty} T_n''(x) X_n(x) + a^2 \sum_{n=0}^{\infty} T_n(x) [\lambda_n X_n(x)] = \sum_{n=0}^{\infty} [T_n''(x) + a^2 \lambda_n T_n(x)] X_n(x) = 0$

$$\Rightarrow T_n''(x) + a^2 \lambda_n T_n(x) = 0$$

常微分方程初始条件

$$\begin{cases} u(x, 0) = \sum_{n=0}^{\infty} T_n(0) X_n(x) = x \\ u_t(x, 0) = \sum_{n=0}^{\infty} T_n'(0) X_n(x) = 0 \end{cases} \Rightarrow \begin{cases} T_n''(x) + a^2 \lambda_n T_n(x) = 0 \\ T_n(t) = \varphi(x), T_n'(t) = 0 \end{cases}$$

$$\varphi_0 = \frac{l}{2}, \varphi_n = \frac{2}{l} \int_0^l x \cos(\frac{n\pi x}{l}) dx = \frac{2l}{(n\pi)^2} [(1-1)^n - 1]$$

$$n=0, T_0(t) = C_{10} + C_{20}t, T_0'(t) = C_{20}$$

$$n \geq 1, T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t), T_n'(t) = a\sqrt{\lambda_n} [-C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t)]$$

$$\begin{cases} T_0(0) = C_{10} = \varphi_0 \\ T_0'(0) = C_{20} = 0 \end{cases}, \begin{cases} T_n(0) = C_{1n} = \varphi_n \\ T_n'(0) = a\sqrt{\lambda_n} C_{2n} = 0 \end{cases}$$

$$\Rightarrow T_n = \varphi_n \cos \frac{n\pi a}{l} t$$

$$\Rightarrow u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} \varphi_n \cos \frac{n\pi a}{l} t \cos \frac{n\pi x}{l}, \text{其中 } n=0, \varphi_0 = \frac{l}{2}, n \geq 1, \varphi_n = \frac{2l}{(n\pi)^2} [(1-1)^n - 1]$$

(2) 设  $u(x, t) = X(x) T(t)$

$$\Rightarrow X(x) T''(t) - a^2 X''(x) T(t) = 0 \Rightarrow \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$\Rightarrow \begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + a^2 \lambda T(t) = 0 \end{cases}$$

齐次边界  $T(t)$  不恒为 0  $\Rightarrow X(0) = X(l) = 0$

特征值:  $\lambda_n = [\frac{(2n+1)\pi}{2l}]^2$ , 特征方程:  $X_n(x) = \sin \frac{(2n+1)\pi}{2l} x$ ,  $n=0, 1, 2, \dots$

分离变量的形式解:  $u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(2n+1)\pi}{2l} x$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=0}^{\infty} T_n(t) \lambda_n X_n(x) = \sum_{n=0}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0 \Rightarrow T_n''(t) + a^2 \lambda_n T_n(t) = 0$$

$$\text{设 } T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t), T_n'(t) = a\sqrt{\lambda_n} [-C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t)]$$

常微分方程初始条件

$$\begin{cases} u(x, 0) = \sum_{n=0}^{\infty} T_n(0) X_n(x) = \sum_{n=0}^{\infty} \varphi_n \\ u_t(x, 0) = \sum_{n=0}^{\infty} T_n'(0) X_n(x) = \sum_{n=0}^{\infty} \psi_n \end{cases} \Rightarrow \begin{cases} \varphi_n = \frac{2}{l} \int_0^l \sin \frac{5\pi}{2l} x \sin \frac{(2n+1)\pi}{2l} x dx = \begin{cases} 1, n=1 \\ 0, \text{其他} \end{cases} \\ \psi_n = \frac{2}{l} \int_0^l \sin \frac{5\pi}{2l} x \sin \frac{(2n+1)\pi}{2l} x dx = \begin{cases} 1, n=2 \\ 0, \text{其他} \end{cases} \end{cases}$$

$$\begin{cases} T_n(0) = C_{1n} = \varphi_n \\ T_n'(0) = a\sqrt{\lambda_n} C_{2n} = \psi_n \end{cases} \Rightarrow C_{11} = 1, C_{22} = \frac{2l}{5\pi a}$$

$$\Rightarrow u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \cos \frac{3\pi a}{2l} t \sin \frac{5\pi}{2l} x + \frac{2l}{5\pi a} \sin \frac{5\pi a}{2l} t \sin \frac{5\pi}{2l} x$$

(3) 设  $u(x, t) = X(x) T(t)$

$$\text{对应的齐次偏微分方程: } T''(t) X(t) - a^2 T(t) X''(t) + 4 T(t) X(t) = 0$$

$$\text{对应的特征值问题: } X''(x) + \lambda X(x) = 0$$

边界条件:

齐次边界  $T(t)$  不恒为 0  $X(0) = X(l) = 0$

特征值:  $\lambda_n = (n\pi)^2$ , 特征函数:  $X_n(x) = \sin n\pi x$ ,  $n=1, 2, 3, \dots$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} X_n(x) T_n(t)$$

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n \lambda_n X_n(x) + 4 \sum_{n=1}^{\infty} T_n(t) X_n(x) = 0 \Rightarrow T_n''(t) + (a^2 \lambda_n + 4) T_n(t) = 0$$

$$\therefore \text{设 } T_n = C_{1n} \cos(\sqrt{a^2 \lambda_n + 4} t) + C_{2n} \sin(\sqrt{a^2 \lambda_n + 4} t)$$

$$T_n' = \sqrt{a^2 \lambda_n + 4} (-C_{1n} \sin(\sqrt{a^2 \lambda_n + 4} t) + C_{2n} \cos(\sqrt{a^2 \lambda_n + 4} t))$$

常微分方程初始条件

$$\begin{cases} u(x, 0) = \sum_{n=1}^{\infty} X_n(x) T_n(0) = x^2 - x = \sum_{n=1}^{\infty} \varphi_n \Rightarrow T_n(0) = \varphi_n, T_n'(0) = 0 \\ u_t(x, 0) = \sum_{n=1}^{\infty} X_n(x) T_n'(0) = 0 \end{cases}$$

$$\varphi_n = 2 \int_0^1 (x^2 - x) \sin n\pi x dx = \frac{8}{(n\pi)^3} [(-1)^n - 1]$$

$$\begin{cases} T_n(0) = C_{1n} = \varphi_n \\ T_n'(0) = \sqrt{a^2 \lambda_n + 4} C_{2n} = 0 \end{cases} \Rightarrow T_n(t) = \varphi_n \cos(\sqrt{n^2 \pi^2 + 4} t)$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \varphi_n \cos(\sqrt{n^2 \pi^2 + 4} t) \sin(n\pi x), \text{ 其中 } \varphi_n = \frac{8}{(n\pi)^3} [(-1)^n - 1]$$

$$(4) \text{ 设 } u(x, t) = X(x) T(t)$$

对应的齐次偏微分方程:  $T''(t)X(x) - T(t)X''(x) - 4T(t)X(x) = 0$

$$\text{对应的特征值问题 } \begin{cases} X'(1) + \lambda X(1) = 0 \\ X(0) = 0, X'(\pi) = 0 \end{cases}$$

特征值:  $\lambda_n = n^2$ , 特征函数:  $X_n(x) = \cos nx, n = 0, 1, 2, \dots$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - T_n(t) X_n''(x) - 4T_n(t) X_n(x) = \sum_{n=0}^{\infty} [T_n''(t) + (\lambda_n - 4)T_n(t)] X_n(x) = \sum_{n=0}^{\infty} f_n X_n$$

$$n > 0, f_n = \frac{2}{\pi} \int_0^{\pi} 2\sin^2 x \cos nx dx = \begin{cases} -1, & n=2 \\ 0, & \text{其他} \end{cases}, n=0, f_0 = \frac{2}{2\pi} \int_0^{\pi} 2\sin^2 x dx = 1$$

$$T_n''(t) + (\lambda_n - 4)T_n(t) = f_n$$

$$n=0 \text{ 时, } T_0'' - 4T_0(t) = 1 \Rightarrow \text{设 } T_0 = C_1 e^{-2t} + C_2 e^{2t} - \frac{1}{4}$$

$$T_0'(t) = -2C_1 e^{-2t} + 2C_2 e^{2t}$$

$$\begin{cases} T_0(0) = C_1 + C_2 - \frac{1}{4} = 0 \\ T_0'(0) = -2C_1 + 2C_2 = 0 \end{cases} \Rightarrow C_1 = C_2 = \frac{1}{8} \Rightarrow T_0 = \frac{1}{8}(e^{-2t} + e^{2t} - 2)$$

$$n=2 \text{ 时, } T_1'' = -1 \Rightarrow \begin{cases} T_1(0) = 0 \\ T_1'(0) = 0 \end{cases} \Rightarrow T_1 = -\frac{t^2}{2}$$

$$n = \text{其他时}, T_n'' + (n^2 - 4)T_n(t) = 0$$

$$n=1, \text{ 设 } T_1 = C_1 e^{-kt} + C_2 e^{kt}, T_1' = k(-C_1 e^{-kt} + C_2 e^{kt})$$

$$\begin{cases} T_1 = C_1 + C_2 = 0 \\ T_1' = k(-C_1 + C_2) = 0 \end{cases} \Rightarrow C_1 = C_2 = 0, T_1(t) = 0$$

$$n \geq 3, \text{ 设 } T_n = C_1 \cos(\sqrt{n^2 - 4} t) + C_2 \sin(\sqrt{n^2 - 4} t)$$

$$T_n'(t) = \sqrt{n^2 - 4} (-C_1 \sin(\sqrt{n^2 - 4} t) + C_2 \cos(\sqrt{n^2 - 4} t)) = 0$$

$$\begin{cases} T_n(0) = C_1 = 0 \\ T_n'(0) = \sqrt{n^2 - 4} C_2 = 0 \end{cases} \Rightarrow C_1 = C_2 = 0, T_n(t) = 0$$

$$\Rightarrow u(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) = \frac{1}{8}(e^{-2t} + e^{2t} - 2) - \frac{t^2}{2} \cos x$$

$$(5) \text{ 设 } u(x, t) = X(x) T(t)$$

对应的齐次偏微分方程:  $T''(t)X(x) - T(t)X''(x) = 0$

$$\text{对应的特征值问题: } \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$$

特征值:  $\lambda_n = (\frac{2n+1}{2l}\pi)^2$ , 特征函数:  $X_n(x) = \sin(\frac{2n+1}{2l}\pi x), n = 0, 1, 2, \dots$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) + \lambda_n T_n(t) X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x) \Rightarrow T_n''(t) + \lambda_n T_n(t) = f_n$$

$$f_n = \frac{2}{l} \int_0^l 2 \sin(\frac{2n+1}{2l}\pi x) dx = \frac{8}{(2n+1)\pi}$$

$$\begin{aligned} \text{设 } T_n &= C_n \cos \frac{2n+1}{2L} \pi t + C_{2n} \sin \frac{2n+1}{2L} \pi t + \frac{8}{(2n+1)\pi} / \left( \cos \frac{2n+1}{2L} \pi \right)^2 \\ &= C_n \cos \left( \frac{2n+1}{2L} \pi t \right) + C_{2n} \sin \left( \frac{2n+1}{2L} \pi t \right) + \frac{32L}{[(2n+1)\pi]^3} \end{aligned}$$

$$T_n' = \frac{2n+1}{2L} \pi [-C_n \sin \frac{2n+1}{2L} \pi t + C_{2n} \cos \frac{2n+1}{2L} \pi t]$$

$$\varphi_n = \frac{2}{L} \int_0^L A \sin \frac{2n+1}{2L} \pi x dx = \frac{4A}{(2n+1)\pi}$$

$$\begin{cases} T_n(0) = C_n + \frac{32L}{[(2n+1)\pi]^3} = 0 \\ T_n'(0) = \frac{2n+1}{2L} \pi C_{2n} = \varphi_n \end{cases} \Rightarrow \begin{cases} C_n = -\frac{32L}{[(2n+1)\pi]^3} \\ C_{2n} = \frac{8AL}{[(2n+1)\pi]^2} \end{cases} \Rightarrow T_n = \frac{32L}{[(2n+1)\pi]^3} \left[ 1 - \cos \frac{2n+1}{2L} \pi t \right] + \frac{8AL}{[(2n+1)\pi]^2} \sin \frac{2n+1}{2L} \pi t$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t)$$

$$\begin{aligned} &= \sum_{n=0}^{\infty} \left( \frac{32L}{[(2n+1)\pi]^3} \left[ 1 - \cos \frac{2n+1}{2L} \pi t \right] + \frac{8AL}{[(2n+1)\pi]^2} \sin \frac{2n+1}{2L} \pi t \right) \sin \frac{2n+1}{2L} \pi x \\ &= 2x \left( 1 - \frac{x}{2} \right) + \sum_{n=0}^{\infty} \left[ -\frac{32L^2}{(2n+1)^3 \pi^3} \cos \frac{(2n+1)}{2L} \pi t + \frac{8AL}{(2n+1)^2 \pi^2} \sin \frac{(2n+1)}{2L} \pi t \right] \pi x \end{aligned}$$

6.

1) 齐次化边界  $w(x, t) = (x - \pi) + \pi = x$   
 $w_t = 0, w_{xx} = 0, w(x, 0) = x \Rightarrow \begin{cases} V_t - a^2 V_{xx} = \cos \frac{x}{2}, 0 < x < \pi, t > 0 \\ V_x(0, t) = 0, V(\pi, t) = 0, t \geq 0 \\ V(x, 0) = -x, 0 \leq x \leq \pi \end{cases}$

设  $V(x, t) = X(x) T(t)$

特征值问题:  $X''(x) + \lambda_n X_n(x) = 0, X'(0) = X(\pi) = 0$

特征值:  $\lambda_n = \left( \frac{2n+1}{2} \right)^2$ , 特征函数:  $X_n(x) = \cos \frac{2n+1}{2} x, n = 0, 1, 2, \dots$

$$\sum_{n=0}^{\infty} T_n'(t) X_n(x) + a^2 \sum_{n=0}^{\infty} \lambda_n X_n(x) T_n(t) = \sum_{n=0}^{\infty} f_n X_n(x) \Rightarrow T_n'(t) + a^2 \lambda_n T_n(t) = f_n$$

$$f_n = \frac{2}{\pi} \int_0^{\pi} \cos \frac{x}{2} \cos \frac{2n+1}{2} x dx = \frac{4}{\pi} \left[ (-1)^{n+1} + \frac{2}{(2n+1)\pi} \right]$$

$\begin{matrix} 2n+1 \\ 0, \text{其他} \end{matrix}$

$$\varphi_n = \frac{2}{\pi} \int_0^{\pi} -x \cos \frac{2n+1}{2} x dx =$$

$$n=0, T_0'(t) + \frac{a^2}{4} T_0(t) = 1 \Rightarrow T_0(t) = C_0 e^{-\frac{a^2}{4} t} + \frac{4}{a^2}$$

$$T_0(0) = C_0 + \frac{4}{a^2} = \varphi_0 \Rightarrow C_0 = \varphi_0 - \frac{4}{a^2} \Rightarrow T_0(t) = \left( \varphi_0 - \frac{4}{a^2} \right) e^{-\frac{a^2}{4} t} + \frac{4}{a^2}$$

$$n \neq 0, T_n'(t) + a^2 \lambda_n T_n(t) = 0 \Rightarrow T_n(t) = C_n e^{-\frac{a^2}{4} t}$$

$$T_n(t) = C_n = \varphi_n \Rightarrow T_n(t) = \varphi_n e^{-a^2 \lambda_n t}$$

$$\begin{aligned} \Rightarrow u(x, t) &= w(x, t) + \sum_{n=0}^{\infty} X_n(x) T_n(t) \\ &= x + \left[ \left( \varphi_0 - \frac{4}{a^2} \right) e^{-\frac{a^2}{4} t} + \frac{4}{a^2} \right] \cos \frac{x}{2} + \sum_{n=1}^{\infty} \varphi_n e^{-a^2 \lambda_n t} \cos \frac{2n+1}{2} x \end{aligned}$$

其中  $\lambda_n = \left( \frac{2n+1}{2} \right)^2, \varphi_n = \frac{4}{\pi} \left[ (-1)^{n+1} + \frac{2}{(2n+1)\pi} \right]$

(2) 设  $u(x, t) = X(x) T(t)$

特征值问题  $X_n''(x) + \lambda_n X_n(x) = 0, X'(0) = X(1) = 0$

特征值:  $\lambda_n = \left( \frac{2n+1}{2} \pi \right)^2, X_n = \cos \frac{2n+1}{2} \pi x, n = 0, 1, 2, \dots$

$$\sum_{n=0}^{\infty} [T_n'(t) + a^2 \lambda_n T_n(t) - 2T_n(t)] X_n(x) = 0$$

$$\Rightarrow T_n'(t) + (a^2 \lambda_n - 2) T_n(t) = 0 \Rightarrow T_n = C_n e^{(2 - a^2 \lambda_n) t}$$

$$T_n(0) = C_n = \varphi_n \therefore T_n = \varphi_n e^{(2 - a^2 \lambda_n) t}$$

$$\varphi_n = 2 \int_0^1 \sin \pi x \cos \frac{2n+1}{2} \pi x dx = \frac{2}{(2n+3)\pi} - \frac{2}{(2n-1)\pi}$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) = \sum_{n=0}^{\infty} \varphi_n e^{(2 - a^2 \lambda_n) t} \cos \frac{2n+1}{2} \pi x$$

其中  $\varphi_n = \frac{2}{(2n+3)\pi} - \frac{2}{(2n-1)\pi}, \lambda_n = \left( \frac{2n+1}{2} \pi \right)^2$

(3) 设  $u(x, t) = X(x) T(t)$

特征值问题:  $X_n''(x) + \lambda_n X_n(x) = 0, X(0) = X(1) = 0$

特征值:  $\lambda_n = \left( \frac{n\pi}{L} \right)^2$ , 特征函数:  $X_n(x) = \sin \left( \frac{n\pi}{L} x \right), n = 1, 2, 3, \dots$

$$T_n'(t) + (a^2 \lambda_n + b^2) T_n(t) = 0 \Rightarrow \text{设 } T_n(t) = C_n e^{-(a^2 \lambda_n + b^2) t}$$

$$T_n(0) = \varphi_n \Rightarrow T_n(t) = \varphi_n e^{-(a^2 \lambda_n + b^2) t}$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} \varphi_n e^{-(a^2 \lambda_n + b^2) t} \sin \frac{n\pi}{L} x$$

其中,  $\lambda_n = \left( \frac{n\pi}{L} \right)^2, \varphi_n = \frac{2}{L} \int_0^L \varphi(x) \sin \frac{n\pi}{L} x dx$

(4) 设  $u(x,t) = X(x)T(t)$

特征值问题:  $X''(x) + \lambda_n X(x) = 0, X_n'(0) = X_n'(l) = 0$

特征值:  $\lambda_n = (\frac{n\pi}{l})^2, X_n(x) = C_1 \cos \frac{n\pi}{l} x, n=0,1,2,\dots$

$T_n'(t) + a^2 \lambda_n T_n(t) = f_n$

$n=0, f_0 = \frac{2}{2l} \int_0^l x l dx = \frac{1}{2}l, \varphi_0 = \frac{2}{2l} \int_0^l dx = 1$

$\Rightarrow T_0'(t) = f_0 \cdot t$  设  $T_0(t) = \frac{1}{4} l t^2 + C_0$

$T_0(0) = C_0 = \varphi_0 = 1 \Rightarrow T_0(t) = 1 + \frac{1}{4} l t^2$

$n \neq 0, f_n = \frac{2}{l} \int_0^l x \cos \frac{n\pi}{l} x dx = \frac{2l}{n^2 \pi^2} [(-1)^n - 1], \varphi_n = \frac{2}{l} \int_0^l \cos \frac{n\pi}{l} x dx = 0$

设  $T_n(t) = C_n e^{-a^2 \lambda_n t} + \frac{f_n}{a^2 \lambda_n} t + \bar{C}_n$

$T_n'(t) = -a^2 \lambda_n C_n e^{-a^2 \lambda_n t} + \frac{f_n}{a^2 \lambda_n} \Rightarrow T_n'(t) + a^2 \lambda_n T_n(t) = f_n \cdot t + a^2 \lambda_n \bar{C}_n + \frac{f_n}{a^2 \lambda_n} = f_n \cdot t$

$\Rightarrow \bar{C}_n = -\frac{f_n}{a^2 \lambda_n}, T_n(0) = C_n - \frac{f_n}{a^2 \lambda_n} \Rightarrow C_n = \frac{f_n}{a^2 \lambda_n^2}$

$\therefore T_n(t) = \frac{f_n}{a^2 \lambda_n^2} (e^{-a^2 \lambda_n t} + a^2 \lambda_n t - 1)$

$\therefore u(x,t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = 1 + \frac{1}{4} l t^2 + \sum_{n=1}^{\infty} [\frac{f_n}{a^2 \lambda_n^2} (e^{-a^2 \lambda_n t} + a^2 \lambda_n t - 1)] \cos \frac{n\pi}{l} x$

其中,  $\lambda_n = (\frac{n\pi}{l})^2, f_n = \frac{2l}{n^2 \pi^2} [(-1)^n - 1]$

7.

(1) 设  $u(x,y) = X(x)Y(y)$ , 令  $\frac{Y''(y)}{Y(y)} = -\frac{X''(x)}{X(x)} = -\lambda$

对应的特征问题:  $\begin{cases} Y''(y) + \lambda_n Y(y) = 0 \\ Y'(0) = Y'(b) = 0 \end{cases}$

$\therefore$  特征值:  $\lambda_n = (\frac{n\pi}{b})^2$ , 特征函数:  $Y_n(y) = C_1 \cos \frac{n\pi}{b} y, n=0,1,2,\dots$

$f_n = \frac{\int_0^b x \cos \frac{n\pi}{b} y dy}{\int_0^b [\cos \frac{n\pi}{b} y]^2 dy} = \begin{cases} x, n=0 \\ 0, n \neq 0 \end{cases}, \varphi_n = \frac{\int_0^b A y \cos \frac{n\pi}{b} y dy}{\int_0^b [\cos \frac{n\pi}{b} y]^2 dy} = \begin{cases} \frac{Ab}{2}, n=0 \\ \frac{2Ab}{(n\pi)^2} [(-1)^n - 1] \end{cases}$

$X_n'' - \lambda_n X_n = f_n$

$n=0, X_0'' = x$  设  $X_0(x) = \frac{1}{6} x^3 + C_1 x + C_0$

$\begin{cases} X_0(0) = C_0 = 0 \\ X_0(a) = \frac{1}{6} a^3 + C_1 a + C_0 = \frac{Ab}{2} \end{cases} \Rightarrow \begin{cases} C_1 = (\frac{Ab}{2a} - \frac{1}{6} a^2) \\ C_0 = 0 \end{cases} \therefore T_0(x) = \frac{1}{6} x^3 + (\frac{Ab}{2a} - \frac{1}{6} a^2) x$

$n \neq 0, X_n'' - \lambda_n X_n = 0$  设  $X_n(x) = C_{1n} \cosh \frac{n\pi}{b} x + C_{2n} \sinh \frac{n\pi}{b} x$

$\begin{cases} X_n(0) = C_{1n} = 0 \\ X_n(a) = C_{1n} \cosh \frac{n\pi}{b} a + C_{2n} \sinh \frac{n\pi}{b} a = \frac{2Ab}{(n\pi)^2} [(-1)^n - 1] \end{cases} \Rightarrow \begin{cases} C_{1n} = 0 \\ C_{2n} = \frac{2Ab}{(n\pi)^2 \sinh \frac{n\pi}{b} a} [(-1)^n - 1] \end{cases}$

$\therefore X_n(x) = \frac{2Ab}{(n\pi)^2 \sinh \frac{n\pi}{b} a} [(-1)^n - 1] \sinh \frac{n\pi}{b} x$

$\therefore u(x,t) = \sum_{n=0}^{\infty} X_n(x) Y_n(y)$

$= \frac{1}{6} x^3 + (\frac{Ab}{2a} - \frac{1}{6} a^2) x + \sum_{n=1}^{\infty} \left( \frac{2Ab}{(n\pi)^2 \sinh \frac{n\pi}{b} a} [(-1)^n - 1] \sinh \frac{n\pi}{b} x \right) \cos \frac{n\pi}{b} y$

(2) 极坐标化

$\begin{cases} U_{\rho\rho} + \frac{1}{\rho} U_{\rho} + \frac{1}{\rho^2} U_{\theta\theta} = 0, 0 < \theta < \frac{\pi}{4}, 0 < \rho < 2 \\ U(\rho,0) = 0, U(\rho, \frac{\pi}{4}) = 0, 0 \leq \rho \leq 2 \\ U(2,\theta) = 2(\cos\theta + \sin\theta), 0 \leq \theta \leq \frac{\pi}{4} \end{cases}$

设  $u(\rho,\theta) = R(\rho) \cdot \Phi(\theta)$

对应的特征值问题:  $\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(\frac{\pi}{4}) = 0 \end{cases}$

特征值:  $\lambda_n = (4n)^2$ , 特征函数:  $\Phi_n(\theta) = \sin 4n\theta, n=1,2,3,\dots$

$\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0$

令  $\rho = e^s \Rightarrow R''(s) - \lambda R(s) = 0 \Rightarrow$  设  $R(s) = C_{1n} e^{4ns} + C_{2n} e^{-4ns}$

$\because |R(0)| < \infty \therefore C_{2n} = 0 \therefore R(s) = C_{1n} e^{4ns}, \text{ 即 } R(\rho) = C_{1n} \rho^{4n}$

$\varphi_n = \frac{2}{\pi/4} \int_0^{\pi/4} 2(\cos\theta + \sin\theta) \sin 4n\theta d\theta = \frac{128n}{\pi} \frac{[1 - \sqrt{2}(-1)^n]}{32n^2 - 2}$

$$R(2) = 2^{4n} C_{1,n} = \varphi_n \Rightarrow C_{1,n} = \frac{\varphi_n}{2^{4n}} \therefore R(\rho) = \left(\frac{\rho}{2}\right)^{4n} \varphi_n$$

$$\therefore u(\rho, \theta) = \sum_{n=1}^{\infty} R(\rho) \psi_n(\theta) = \sum_{n=1}^{\infty} \varphi_n \left(\frac{\rho}{2}\right)^{4n} \sin 4n\theta, \text{ 其中 } \varphi_n = \frac{128n}{\pi} \frac{[1 - \sqrt{2}(-1)^n]}{32n^2 - 2}$$

(3) 极坐标化: 
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, 0 < \theta < 2\pi \\ u(2, \theta) = 1 + 2\cos\theta, & 0 \leq \theta \leq 2\pi \end{cases}$$

设  $u(\rho, \theta) = R(\rho) \psi(\theta)$   
对应的特征值问题: 
$$\begin{cases} \psi''(\theta) + \lambda \psi(\theta) = 0 \\ \psi(\theta) = \psi(2\pi + \theta) \end{cases}$$

特征值:  $\lambda_n = n^2$ , 特征函数:  $\psi(\theta) = a_n \cos n\theta + b_n \sin n\theta$   
 $\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0 \Rightarrow R_0 = C_0; R_n = C_n \rho^n, n \geq 1$   
 $u(\rho, \theta) = C_0 + \sum_{n=1}^{\infty} C_n \rho^n \psi_n(\theta)$  代入  $u(2, \theta) = 1 + 2\cos\theta$   
即  $C_0 + \sum_{n=1}^{\infty} 2^n \cdot C_n (a_n \cos n\theta + b_n \sin n\theta) = 1 + 2\cos\theta$

比较系数法 
$$\begin{cases} C_0 = 1 \\ C_1 a_1 = 1, C_1 b_1 = 0 \\ C_n a_n = C_n b_n = 0, n \geq 2 \end{cases} \Rightarrow u(\rho, \theta) = 1 + \rho \cos\theta$$

(4) 极坐标化: 
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = \frac{1}{2} \rho^2 \sin 2\theta, & 1 < \rho < 2, 0 < \theta < 2\pi \\ u(1, \theta) = 0, u(2, \theta) = 2(\cos\theta + \sin\theta), & 0 \leq \theta \leq 2\pi \end{cases}$$

设  $u(\rho, \theta) = R(\rho) \psi(\theta)$   
对应的特征值问题: 
$$\begin{cases} \psi''(\theta) + \lambda \psi(\theta) = 0 \\ \psi(\theta) = \psi(\theta + 2\pi) \end{cases}$$

特征值:  $\lambda_n = n^2$ , 特征函数:  $\psi(\theta) = \{1, \cos n\theta, \sin n\theta\}, n = 0, 1, 2, \dots$   
 $u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \psi_n(\theta)$   
即  $\frac{1}{2} \rho^2 \sin 2\theta = \sum_{n=0}^{\infty} f_{n,\cos}(\rho) \cos n\theta + \sum_{n=0}^{\infty} f_{n,\sin}(\rho) \sin n\theta$   
比较系数法: 
$$\begin{cases} f_{0,\cos}(\rho) = \epsilon_0 + D_0 \ln \rho, f_{n,\cos}(\rho) = \epsilon_n \rho^n + D_n \rho^{-n}, n \geq 1 \\ f_{0,\sin}(\rho) = c_0 + d_0 \ln \rho, f_{n,\sin}(\rho) = C_n \rho^n + d_n \rho^{-n}, n \geq 1, n \neq 2 \\ f_{2,\sin}(\rho) = C' + d \rho^{-2} + \frac{1}{24} \rho^4 \end{cases}$$
  
 $u(1, \theta) = \sum_{n=0}^{\infty} f_{n,\cos}(1) \cos n\theta + \sum_{n=0}^{\infty} f_{n,\sin}(1) \sin n\theta = 0 \therefore f_{n,\cos}(1) = f_{n,\sin}(1) = 0$   
 $u(2, \theta) = \sum_{n=0}^{\infty} f_{n,\cos}(2) \cos n\theta + \sum_{n=0}^{\infty} f_{n,\sin}(2) \sin n\theta = 2(\cos\theta + \sin\theta) \therefore \begin{cases} f_{1,\cos}(2) = 2, f_{1,\sin}(2) = 2 \\ f_{n,\cos}(2) = f_{n,\sin}(2) = 0, n \neq 1 \end{cases}$   
 $\therefore \begin{cases} \epsilon_n = D_n = C_n = d_n = 0, n \geq 2 \\ \epsilon_1 = \frac{4}{3}, D_1 = -\frac{4}{3}, C_1 = \frac{4}{3}, d_1 = -\frac{4}{3} \\ C = -\frac{7}{45}, d = \frac{2}{15} \end{cases}$

$$\therefore u(\rho, \theta) = \frac{4}{3} (\rho - \rho^{-1}) (\sin\theta + \cos\theta) + \left(-\frac{7}{45} \rho^2 + \frac{2}{15} \rho^{-2} + \frac{1}{24} \rho^4\right) \sin 2\theta$$

补充题 
$$\begin{cases} u_{xx} + u_{yy} = 0, x > 0, y > 0, x^2 + y^2 < 4 \\ u(x, 0) = 0, 0 \leq x \leq 1 \\ u(0, y) = 0, 0 \leq y \leq 2 \\ u(x, y) = xy, x^2 + y^2 = 4 \end{cases}$$
 极坐标化 
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = u(\rho, \frac{\pi}{2}) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

设  $u(\rho, \theta) = R(\rho) \psi(\theta)$   
对应的特征方程 
$$\begin{cases} \psi''(\theta) + \lambda \psi(\theta) = 0 \\ \psi(0) = \psi(\frac{\pi}{2}) = 0 \end{cases}$$

特征值:  $\lambda_n = (2n)^2$ , 特征函数:  $\psi_n(\theta) = \sin 2n\theta, n = 1, 2, 3, \dots$   
 $\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0$  设  $\rho = e^s, R'(s) - \lambda R(s) = 0$   
 $\therefore R_n(s) = C_{1n} e^{2ns} + C_{2n} e^{-2ns}$  即  $R_n(\rho) = C_{1n} \rho^{2n} + C_{2n} \rho^{-2n} \therefore |R_n(0)| < \infty \therefore C_{2n} = 0$   
 $\therefore u(\rho, \theta) = \sum_{n=1}^{\infty} R_n(\rho) \psi_n(\theta) = \sum_{n=1}^{\infty} C_{1n} \rho^{2n} \sin 2n\theta, u(2, \theta) = \sum_{n=1}^{\infty} 2^{2n} C_{1n} \sin 2n\theta = 2 \sin 2\theta$

比较系数法:  $C_{1n} = \begin{cases} 0, n \neq 1 \\ \frac{1}{2}, n = 1 \end{cases} \therefore u(\rho, \theta) = \frac{1}{2} \rho^2 \sin 2\theta$  即  $u(x, y) = xy$

8.

(1)  $C_k = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{k\pi}{l} x dx$

$\int_0^l \varphi(x) S_n(x) = \int_0^l \varphi(x) \sum_{k=1}^n C_k \sin \frac{k\pi}{l} x dx = \sum_{k=1}^n C_k \int_0^l \varphi(x) \sin \frac{k\pi}{l} x dx = \frac{1}{2} \sum_{k=1}^n C_k^2$

正交性:  $\int_0^l S_n^2(x) dx = \int_0^l \sum_{k=1}^n C_k^2 \sin^2 \frac{k\pi}{l} x dx = \frac{1}{2} \sum_{k=1}^n C_k^2$

$\int_0^l [\varphi(x) - S_n(x)]^2 = \int_0^l \varphi^2(x) dx - \frac{1}{2} \sum_{k=1}^n C_k^2$

(2)  $[\varphi(x) - S_n(x)]^2 \geq 0 \therefore \int_0^l \varphi^2(x) dx - \frac{1}{2} \sum_{k=1}^n C_k^2 = \int_0^l [\varphi(x) - S_n(x)]^2 \geq 0$

移项  $\sum_{k=1}^n C_k^2 \leq \frac{2}{l} \int_0^l \varphi^2(x) dx$  令  $n \rightarrow \infty$ , 则  $\sum_{k=1}^{\infty} C_k^2 \leq \frac{2}{l} \int_0^l \varphi^2(x) dx$ , 证毕

(3)  $d_n = \frac{2}{l} \int_0^l \varphi'(x) \sin \frac{n\pi}{l} x dx = \frac{2}{l} ([\varphi(x) \sin \frac{n\pi}{l} x]_0^l - \frac{n\pi}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx)$   
 $= -\frac{2n\pi}{l^2} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx$   
 $= -\frac{2n\pi}{l^2} ([\varphi(x) \cos \frac{n\pi}{l} x]_0^l + \frac{n\pi}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx)$

$\varphi(0) = \varphi(l) = 0$

$d_n = -(\frac{n\pi}{l})^2 (\frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx) = -(\frac{n\pi}{l})^2 C_n \Rightarrow |d_n| = A n^2 |C_n|$ , 其中  $A = (\frac{\pi}{l})^2$

由③可知  $|C_n| = \frac{1}{A n^2} |d_n|$  由②可知:  $\sum_{n=1}^{\infty} d_n^2 \leq \frac{2}{l} \int_0^l \varphi'^2(x) dx$

(4)  $\varphi(x) = \sum_{k=1}^{\infty} C_k \sin \frac{k\pi}{l} x, \varphi'(x) = \sum_{k=1}^{\infty} C_k \frac{k\pi}{l} \cos \frac{k\pi}{l} x$

$|\sin \frac{n\pi}{l} x| \leq 1, |C_n| = A \frac{|d_n|}{n^2}, n \geq 1$

$|C_n| \frac{n\pi}{l} = A \frac{\pi}{l} \frac{|d_n|}{n} \leq \frac{A\pi}{2l} [\frac{1}{n^2} + d_n^2], n \geq 1$

M-判别法: 所求两级数在区间  $[0, l]$  上一致收敛

9.

(1) 一根长度为  $l$  的均匀导热杆的热传导过程

(2)  $\because U_x(0, t) = U_x(l, t) = 0$ , 即杆两端绝热, 充分长时间后, 温度趋于均匀

(3)  $\because$  两端绝热

$\therefore$  系统总热量守恒  $\therefore \lim_{t \rightarrow +\infty} U(x, t) = \frac{1}{l} \int_0^l \varphi(x) dx$

10.

(1) 一根长度为  $l$  的均匀导热杆的热传导过程

(2)  $\because U(0, t) = A, U_x(l, t) = B$ , 趋于稳态后与时间无关, 设  $\lim_{t \rightarrow +\infty} U(x, t) = v(x)$

代入  $U_{tt} = a^2 U_{xx}$

$v''(x) = 0 \Rightarrow v(x) = C_1 x + C_2$

$\begin{cases} v(0) = C_2 = A \\ v'(l) = C_1 = B \end{cases} \Rightarrow v(x) = A + Bx$  即  $\lim_{t \rightarrow +\infty} U(x, t) = A + Bx$

11.

(1)  $U_{tt}$ : 位移加速度  $U_{xx}$ : 形变加速度

$2U_t$ : 阻尼力  $U$ : 弹性恢复力

(2) 设  $U(x, t) = X(x)T(t)$

代入原方程:  $T''(t)X(x) - T(t)X''(x) + 2T'(t)X(x) + T(t)X(x) = 0$

$\Rightarrow \frac{T''(t) + 2T'(t) + T(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda$

$\because T(t)$  不恒为 0  $\therefore$  特征值问题:  $\begin{cases} X'(x) + \lambda X(x) = 0 \\ X(0) = X(\pi) = 0 \end{cases}$

$\therefore$  特征值:  $\lambda_n = n^2$ , 特征函数:  $X_n(x) = \sin nx, n = 1, 2, 3, \dots$

(3)  $\sum_{n=1}^{\infty} [T_n''(t) + 2T_n'(t) + (1 + \lambda_n)T_n(t)] X_n(x) = 0 \Rightarrow T_n''(t) + 2T_n'(t) + (1 + n^2)T_n(t) = 0$

特征方程:  $r^2 + 2r + (1 + n^2) = 0 \Rightarrow r = -1 \pm ni$

$\therefore$  设  $T_n(t) = e^{-t} (C_{n1} \cos nt + C_{n2} \sin nt) \therefore T_n'(t) = e^{-t} [(-C_{n1} + C_{n2}) \cos nt + (-C_{n1} - C_{n2}) \sin nt]$

$\varphi_n = \frac{2}{\pi} \int_0^{\pi} x \sin nx dx = \frac{2 \times (-1)^{n+1}}{n}, \varphi_1 = 2$

$\begin{cases} T_1(0) = C_{11} = \varphi_1 = 2 \\ T_1'(0) = -C_{11} + C_{12} = 0 \end{cases} \Rightarrow C_{11} = C_{12} = 2 \Rightarrow T_1(t) = 2e^{-t} (\cos t + \sin t)$

12.

(1) 设  $u(x, t) = X(x) T(t)$

对应的齐次偏微分方程:  $T''(t) X(x) - T(t) X''(x) = 0$

对应的特征值问题  $\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(\pi) = 0 \end{cases}$

$\therefore$  特征值:  $\lambda_n = n^2$ , 特征函数:  $X_n(x) = \cos nx, n = 0, 1, 2, \dots$

(2)  $f(x, t) = \sin 10t$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - T_n(t) X_n''(x) = \sum_{n=0}^{\infty} [T_n''(t) + \lambda_n T_n(t)] X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{2}{2\pi} \int_0^\pi \sin 10t dx = \sin 10t$$

$$n \neq 0, f_n = \frac{2}{\pi} \int_0^\pi \sin 10t \cos nx dx = 0$$

$$T_n''(t) + \lambda_n T_n(t) = f_n$$

$n = 0$ , 设  $T_0(t) = -\frac{1}{100} \sin 10t + Ct, T_0'(t) = -\frac{1}{10} \cos 10t + C$

$$\begin{cases} T_0(0) = 0 \\ T_0'(0) = -\frac{1}{10} + C \Rightarrow C = \frac{1}{10} \Rightarrow T_0(t) = \frac{1}{100} (10t - \sin 10t) \end{cases}$$

$n \neq 0$ , 设  $T_n(t) = C_{1n} \cos nt + C_{2n} \sin nt$

$$T_n'(t) = n[-C_{1n} \sin nt + C_{2n} \cos nt]$$

$$\begin{cases} T_n(0) = C_{1n} = 0 \\ T_n'(0) = n C_{2n} = 0 \end{cases} \Rightarrow \begin{cases} C_{1n} = 0 \\ C_{2n} = 0 \end{cases} \Rightarrow T_n(t) = 0$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \frac{1}{100} (10t - \sin 10t)$$

(3)  $u_{tt} = u_{xx} \Rightarrow 2u_{tt} \cdot u_t = 2u_{xx} \cdot u_t$

$$2u_{tt} \cdot u_t = \frac{d}{dt} (u_t^2), 2u_t u_{xx} = 2(u_t u_x)_x - 2u_t x u_x = 2(u_t u_x)_x - \frac{d}{dt} (u_x^2)$$

$$\Rightarrow \frac{d}{dt} (u_t^2 + u_x^2) = \frac{d}{dx} (2u_t u_x)$$

对  $x$  从 0 到  $\pi$  积分  $\int_0^\pi \frac{d}{dt} (u_t^2 + u_x^2) dx = \int_0^\pi \frac{d}{dx} (2u_t u_x) dx = 2u_t u_x \Big|_0^\pi$

代入  $u_x(0, t) = u_x(\pi, t) = 0 \Rightarrow \int_0^\pi \frac{d}{dt} (u_t^2 + u_x^2) dx = \frac{d}{dt} \int_0^\pi u_t^2 + u_x^2 = 0$

$\therefore$  对于任意  $t > 0$ , 积分值等于  $t = 0$  时的值

$$\therefore \int_0^\pi [u_t^2(x, t) + u_x^2(x, t)] dx = \int_0^\pi [u_t^2(x, 0) + u_x^2(x, 0)] dx$$

代入  $u_t(x, 0) = \psi(x), u_x(x, 0) = \varphi_x(x)$

$$\int_0^\pi [u_t^2(x, t) + u_x^2(x, t)] dx = \int_0^\pi [\psi^2(x) + \varphi_x^2(x)] dx$$

$\frac{1}{2} \int u_t^2 dx$  对应系统的动能,  $\frac{1}{2} \int u_x^2 dx$  对应系统的势能

该等式表明能量守恒

(4) 反证法: 设该定解问题存在两个解  $u_1(x, t), u_2(x, t)$

令  $w(x, t) = u_1(x, t) - u_2(x, t)$

$$\Rightarrow \begin{cases} w_{tt} - w_{xx} = 0 \\ w_x(0, t) = w_x(\pi, t) = 0 \\ w(x, 0) = 0, w_t(x, 0) = 0 \end{cases} \Rightarrow \int_0^\pi [w_t^2(x, t) + w_x^2(x, t)] dx = 0$$

$$\Rightarrow w_t(x, t) = w_x(x, t) = 0 \Rightarrow w(x, t) = C$$

又  $\because w(x, 0) = 0 \therefore C = 0 \therefore w(x, t) \equiv 0 \therefore$  只有唯一解

# 习题3

$$1. (1) \text{ 设 } y = \sum_{k=0}^{\infty} a_k x^k, y' = \sum_{k=1}^{\infty} k a_k x^{k-1}, y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$$

$$y'' + x y' = 0 \Rightarrow \sum_{k=1}^{\infty} k a_k x^{k+1} + \sum_{k=2}^{\infty} k(k-1) a_k x^{k-1} = 0$$

$$\text{即 } \sum_{k=2}^{\infty} [(k+1) a_{k+1} + (k+1)(k+2) a_{k+2}] x^k = 0$$

$$a_2 = a_3 = 0$$

$$\therefore a_{k+2} = -\frac{k-1}{(k+1)(k+2)} a_{k-1}$$

$$\therefore a_2 = a_3 = 0 \therefore a_2, 5, 8, \dots = 0, a_3, 6, 9, \dots = 0$$

$$\text{对于 } a_1 \text{ 相关项: } a_{3k+1} = \frac{(-1)^k}{3^k k! (3k+1)} a_1 \therefore y = a_0 + a_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{3^k k! (3k+1)} x^{3k+1}$$

$$(2) \text{ 设 } y = \sum_{k=0}^{\infty} a_k x^k, y' = \sum_{k=1}^{\infty} k a_k x^{k-1}, y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$$

$$y'' + x y' + y = 0 \Rightarrow \sum_{k=0}^{\infty} a_k x^k + \sum_{k=1}^{\infty} k a_k x^k + \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} = 0$$

$$\text{即 } \sum_{k=0}^{\infty} [a_k + k a_k + (k+1)(k+2) a_{k+2}] x^k = 0$$

$$\therefore a_{k+2} = -\frac{1}{k+2} a_k \therefore a_{2k} = \frac{(-1)^k}{(2k)!!} a_0, a_{2k+1} = \frac{(-1)^k}{(2k+1)!!} a_1, k \geq 0$$

$$\therefore y(x) = a_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!!} x^{2k} + a_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!!} x^{2k+1}$$

$$(3) \text{ 设 } y = \sum_{k=0}^{\infty} a_k x^{k+p}, y' = \sum_{k=0}^{\infty} (k+p) a_k x^{k+p-1}, y'' = \sum_{k=0}^{\infty} (k+p)(k+p-1) a_k x^{k+p-2}$$

$$x y'' + y = 0 \Rightarrow \sum_{k=0}^{\infty} a_k x^{k+p} + \sum_{k=0}^{\infty} (k+p)(k+p-1) a_k x^{k+p-1} = 0$$

$$\text{最低项 } x^{p-1}: a_0 p(p-1) = 0 \therefore p = 1 \text{ 或 } 0$$

$$\textcircled{1} p=0 \text{ 时, } a_k \text{ 恒为 } 0$$

$$\textcircled{2} p=1 \text{ 时}$$

$$\therefore a_{k+1} = -\frac{a_k}{k(k+1)} \Rightarrow a_k = \frac{(-1)^k a_0}{k! (k+1)!}, k \geq 0$$

$$\therefore y(x) = a_0 x \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (k+1)!} x^k$$

$$(4) \text{ 设 } y = \sum_{k=0}^{\infty} a_k x^{k+p}, y' = \sum_{k=0}^{\infty} (k+p) a_k x^{k+p-1}, y'' = \sum_{k=0}^{\infty} (k+p)(k+p-1) a_k x^{k+p-2}$$

$$2x y'' + (1-2x) y' - y = 0 \Rightarrow \sum_{k=0}^{\infty} -(2k+2p+1) a_k x^{k+p} + [(k+p) + 2(k+p)(k+p-1)] a_k x^{k+p-1} = 0$$

$$\text{最低项 } x^{p-1}: p(2p-1) a_0 = 0 \therefore p = 0 \text{ 或 } \frac{1}{2}$$

$$\text{又有 } a_{k+1} = \frac{1}{k+p+1} a_k, k \geq 0$$

$$\textcircled{1} p=0 \text{ 时, } a_{k+1} = \frac{1}{k+1} a_k, k \geq 0 \therefore a_k = \frac{1}{k!} a_0 \therefore y_1(x) = a_0 \sum_{k=0}^{\infty} \frac{1}{k!} x^k = a_0 e^x$$

$$\textcircled{2} p=\frac{1}{2} \text{ 时, } a_{k+1} = \frac{2}{2k+3} a_k, k \geq 0 \therefore a_k = \frac{2^k}{(2k+1)!!} a_0 \therefore y_2(x) = a_0 x^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{2^k}{(2k+1)!!} x^k$$

$$\text{显然, } y_1(x), y_2(x) \text{ 线性无关 } \therefore y(x) = \{y_1(x), y_2(x)\}$$

$$2. (1) \text{ 令 } t = x^2 \therefore I = \int_0^{\infty} t^{\frac{5}{4}-1} e^{-t} \frac{1}{2t^{\frac{1}{2}}} dt = \frac{1}{2} \int_0^{\infty} t^{\frac{3}{4}-1} e^{-t} dt = \frac{1}{2} \Gamma(\frac{5}{4}) = \frac{3}{8} \Gamma(\frac{3}{4})$$

$$(2) \text{ 令 } t = \frac{x^2}{2} \therefore I = \int_0^{\infty} 2t e^{-t} \frac{1}{\sqrt{2t}} dt = \sqrt{2} \int_0^{\infty} t^{\frac{3}{2}-1} e^{-t} dt = \sqrt{2} \Gamma(\frac{3}{2}) = \sqrt{2} \Gamma(\frac{1}{2}) = \sqrt{2\pi}$$

$$3. \text{ 已知 } \Gamma(\frac{1}{2}) = \sqrt{\pi} \therefore \Gamma(k - \frac{1}{2} + 1) = \Gamma(k + \frac{1}{2}) = \frac{(2k-1)!!}{2^k} \sqrt{\pi}$$

$$\therefore \int_{-\frac{1}{2}}^{-\frac{1}{2}} = (\frac{x}{2})^{-\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+\frac{1}{2})} (\frac{x}{2})^{2k} = \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (2k-1)!!} \frac{2^k}{2^{2k}} x^{2k}$$

$$\text{又 } (2k)! = 2^k k! (2k-1)!!$$

$$\therefore \int_{-\frac{1}{2}}^{-\frac{1}{2}} = \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \frac{x^{2k}}{2^k} = \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \frac{x^{2k}}{2^k}$$

$$4. x^r J_r(x) = \frac{1}{2^r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} x^{\frac{2k}{2} + r} = \frac{1}{2^r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} x^{2k+r}$$

$$\therefore [x^r J_r(x)]' = \frac{1}{2^r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} x^{2k+r-1} \times \frac{2k+r-1}{2^k}$$

$$= \frac{1}{2^{r-1}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r)} x^{2k+r-1} \times (\frac{x}{2})^{2k} x^{2r-1}$$

$$= x^r [(\frac{x}{2})^{r-1} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r)} (\frac{x}{2})^{2k}]$$

$$= x^r J_{r-1}(x)$$

$$\begin{aligned}
 x^{-r} J_r(x) &= \frac{1}{2^r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k} \\
 [x^{-r} J_r(x)]' &= \frac{1}{2^r} \sum_{k=1}^{\infty} \frac{(-1)^k k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k-1} \\
 &= \frac{1}{2^{r+1}} \sum_{k=0}^{\infty} \frac{(-1)^{k+1} x}{k! \Gamma(k+r+1+1)} \left(\frac{x}{2}\right)^{2k} \\
 &= -x^{-r} \left[\left(\frac{x}{2}\right)^{r+1} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1+1)} \left(\frac{x}{2}\right)^{2k}\right] \\
 &= -x^{-r} J_{r+1}(x)
 \end{aligned}$$

$$5. \therefore J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$$

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

$$\therefore J_{\frac{3}{2}}(x) = \frac{1}{x} J_{\frac{1}{2}}(x) - J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left[ \sin\left(x - \frac{\pi}{2}\right) + \frac{1}{x} \cos\left(x - \frac{\pi}{2}\right) \right]$$

$$6. \therefore 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x),$$

$$\therefore 2J_1'(x) = J_0(x) - J_2(x) \quad \therefore J_1'(x) = \frac{1}{2} [J_0(x) - J_2(x)]$$

$$\text{又} \therefore J_0'(x) = -J_1(x) \quad \therefore J_0'(x) = \frac{1}{2} [J_2(x) - J_0(x)]$$

$$\therefore J_0'''(x) = \frac{1}{2} [J_2(x) - J_0(x)]' = -\frac{1}{2} J_1'(x) + \frac{1}{2} J_1'(x) = -\frac{1}{2} J_0'(x) + \frac{1}{4} [J_1(x) - J_3(x)] = -\frac{1}{4} [3J_0'(x) + J_3(x)]$$

$$\therefore J_3(x) = -3J_0'(x) - 4J_0'''(x)$$

7.

$$(1) r = \frac{2}{5}, \text{ 不为整数} \quad \therefore x = C_1 J_{\frac{2}{5}}(t) + C_2 J_{-\frac{2}{5}}(t)$$

$$(2) r = \sqrt{2}, \text{ 不为整数} \quad \therefore y = C_1 J_{\sqrt{2}}(x) + C_2 J_{-\sqrt{2}}(x)$$

$$8. e^{\frac{x}{2}t} = \sum_{j=0}^{\infty} \frac{(\frac{x}{2}t)^j}{j!} = \sum_{j=0}^{\infty} \frac{x^j}{2^j j!} t^j, \quad e^{-\frac{x}{2t}} = \sum_{k=0}^{\infty} \frac{(-\frac{x}{2t})^k}{k!} = \sum_{k=0}^{\infty} \frac{(-1)^k x^k}{2^k k!} t^{-k}$$

$$\psi(t, x) = e^{\frac{x}{2}t} \cdot e^{-\frac{x}{2t}} = \left[ \sum_{j=0}^{\infty} \frac{x^j}{2^j j!} t^j \right] \cdot \left[ \sum_{k=0}^{\infty} \frac{(-1)^k x^k}{2^k k!} t^{-k} \right]$$

$$\begin{aligned}
 \text{设 } j = k+n, \text{ 对于 } t^n \text{ 项: } a_n &= \sum_{k=0}^{\infty} \left[ \frac{x^{n+k}}{2^{n+k} (n+k)!} \cdot \frac{(-1)^k x^k}{2^k k!} \right] \\
 &= \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \left[ \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k} \right] = J_n(x)
 \end{aligned}$$

$$\therefore \psi(t, x) = \sum_{n=-\infty}^{+\infty} J_n(x) t^n$$

9.

$$(1) \text{ 对于 } \frac{2n}{x} J_n(x) = J_{n-1}(x) + J_{n+1}(x)$$

$$\psi_t = \frac{x}{2} (1+t^{-2}) e^{\frac{x}{2}(t-\frac{1}{t})} = \frac{x}{2} (1+\frac{1}{t^2}) \psi, \quad \psi_t = \sum_{n=-\infty}^{+\infty} n J_n(x) t^{n-1}$$

$$\therefore \sum_{n=-\infty}^{+\infty} n J_n(x) t^{n-1} = \frac{x}{2} (1+t^{-2}) \sum_{n=-\infty}^{+\infty} J_n(x) t^n = \frac{x}{2} \sum_{n=-\infty}^{+\infty} J_n(x) t^n + J_n(x) t^{n-2}$$

$$\text{整理序号: } \sum_{n=-\infty}^{+\infty} n J_n(x) t^{n-1} = \frac{x}{2} \sum_{n=-\infty}^{+\infty} J_{n-1}(x) t^{n-1} + J_{n+1}(x) t^{n-1}$$

$$\therefore n J_n(x) = \frac{x}{2} [J_{n-1}(x) + J_{n+1}(x)] \quad \text{即 } \frac{2n}{x} J_n(x) = J_{n-1}(x) + J_{n+1}(x), \text{ 证毕}$$

$$(2) \text{ 对于 } 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$$

$$\psi_x = \frac{1}{2} (t - \frac{1}{t}) e^{\frac{x}{2}(t-\frac{1}{t})} = \frac{1}{2} (t - \frac{1}{t}) \psi, \quad \psi_x = \sum_{n=-\infty}^{+\infty} J_n'(x) t^n$$

$$\therefore \sum_{n=-\infty}^{+\infty} J_n'(x) t^n = \frac{1}{2} (t - \frac{1}{t}) \sum_{n=-\infty}^{+\infty} J_n(x) t^n = \frac{1}{2} \sum_{n=-\infty}^{+\infty} J_n(x) t^{n+1} - J_n(x) t^{n-1}$$

整理序号:  $\sum_{n=-\infty}^{+\infty} J_n'(x) t^n = \frac{1}{2} \sum_{n=-\infty}^{+\infty} J_{n-1}(x) t^n - J_{n+1}(x) t^n$

$\therefore J_n'(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$  即  $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$ . 证毕

10.

(1) 易知  $J_n(x)$  满足贝塞尔方程:  $x^2 J_n''(x) + x J_n'(x) + (x^2 - n^2) J_n(x) = 0$

$\therefore J_n''(x) = -\frac{1}{x} [J_n'(x) + (x - \frac{n^2}{x}) J_n(x)]$

$\therefore \frac{2n}{x} J_n(x) = J_{n-1}(x) + J_{n+1}(x), 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$

$\therefore J_n''(x) = \frac{1}{2x} [n(J_{n-1}(x) + J_{n+1}(x)) - (J_{n-1}(x) - J_{n+1}(x)) - 2x J_n(x)]$   
 $= \frac{1}{2x} [(n-1)J_{n-1}(x) + (n+1)J_{n+1}(x) - 2x J_n(x)]$

(2) 由题意知:  $J_n'(\alpha_m) = 0$

$J_n''(\alpha_m) = -\frac{1}{\alpha_m^2} [\alpha_m J_n'(\alpha_m) + (\alpha_m^2 - n^2) J_n(\alpha_m)] = \frac{n^2 - \alpha_m^2}{\alpha_m^2} J_n(\alpha_m) = (\frac{n^2}{\alpha_m^2} - 1) J_n(\alpha_m)$

11.

(1)  $\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0$  两边同时除以  $\rho$ :  $(\rho R')' + (\lambda \rho - \frac{n^2}{\rho}) R = 0$

$\therefore \int_0^{\rho_0} R(\rho R')' d\rho + \lambda \int_0^{\rho_0} \rho R^2 d\rho - n^2 \int_0^{\rho_0} \frac{R^2}{\rho} d\rho = 0$

利用分部积分法和  $R'(\rho_0) = 0$ :  $-\int_0^{\rho_0} \rho (R')^2 d\rho + \lambda \int_0^{\rho_0} \rho R^2 d\rho - n^2 \int_0^{\rho_0} \frac{R^2}{\rho} d\rho = 0$

$\therefore \lambda = \frac{\int_0^{\rho_0} [\rho (R')^2 + \frac{n^2}{\rho} R^2]}{\int_0^{\rho_0} \rho R^2 d\rho} \geq 0$   $\lambda$  显然不为 0  $\therefore \lambda > 0$

(2) 易知通解为  $R(\rho) = C_1 J_n(\sqrt{\lambda} \rho) + C_2 N_n(\sqrt{\lambda} \rho)$

$\therefore |R(\rho)| < +\infty \therefore C_2 = 0$

$R'(\rho_0) = C_1 \sqrt{\lambda} J_n'(\sqrt{\lambda} \rho_0) = 0$  设  $\alpha_m$  是  $J_n'(x)$  的正零点,  $m = 1, 2, 3, \dots$

$\therefore \sqrt{\lambda} \rho_0 = \alpha_m \therefore \lambda = (\frac{\alpha_m}{\rho_0})^2, R(\rho) = C_1 J_n(\frac{\rho \alpha_m}{\rho_0}), m = 1, 2, 3, \dots$

(3)  $\begin{cases} (\rho R_m')' + (\lambda_m \rho - \frac{n^2}{\rho}) R_m = 0 \\ (\rho R_k')' + (\lambda_k \rho - \frac{n^2}{\rho}) R_k = 0 \end{cases} \Rightarrow R_k(\rho R_m')' - R_m(\rho R_k')' + (\lambda_m - \lambda_k) \rho R_m R_k = 0$

即  $(\lambda_m - \lambda_k) \rho R_m R_k = R_m(\rho R_k') - R_k(\rho R_m')$

$\therefore \int_0^{\rho_0} (\lambda_m - \lambda_k) \rho R_m R_k d\rho = \int_0^{\rho_0} R_m(\rho R_k') - R_k(\rho R_m') d\rho$   
 $= (\rho R_m R_k')|_0^{\rho_0} - \int_0^{\rho_0} \rho R_m' R_k' d\rho + \int_0^{\rho_0} \rho R_m' R_k' d\rho - (\rho R_k R_m')|_0^{\rho_0}$   
 $= 0$

$\therefore \lambda_m \neq \lambda_k \therefore \int_0^{\rho_0} \rho R_m(\rho) R_k(\rho) d\rho = 0$

(4)  $\int_0^{\rho_0} \rho R_m^2(\rho) d\rho = \frac{\rho_0^2}{2} [(J_n'(\sqrt{\lambda_m} \rho_0))^2 + (1 - \frac{n^2}{(\sqrt{\lambda_m} \rho_0)^2}) J_n^2(\sqrt{\lambda_m} \rho_0)]$   
 $= \frac{\rho_0^2}{2} [0 + (1 - \frac{n^2}{\alpha_m^2}) J_n^2(\alpha_m)]$   
 $= \frac{\rho_0^2}{2} [1 - \frac{n^2}{\alpha_m^2}] J_n^2(\alpha_m)$

12.

(1)  $\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0$  两边同时除以  $\rho$ :  $(\rho R')' + (\lambda \rho - \frac{n^2}{\rho}) R = 0$

$\therefore \int_0^{\rho_0} R(\rho R')' d\rho + \lambda \int_0^{\rho_0} \rho R^2 d\rho - n^2 \int_0^{\rho_0} \frac{R^2}{\rho} d\rho = 0$

利用分部积分法和  $R(\rho_0) = 0$ :  $-\int_0^{\rho_0} \rho (R')^2 d\rho + \lambda \int_0^{\rho_0} \rho R^2 d\rho - n^2 \int_0^{\rho_0} \frac{R^2}{\rho} d\rho = 0$

$\therefore \lambda = \frac{\int_0^{\rho_0} [\rho (R')^2 + \frac{n^2}{\rho} R^2]}{\int_0^{\rho_0} \rho R^2 d\rho} \geq 0$   $\lambda$  显然不为 0  $\therefore \lambda > 0$

(2) 易知通解为  $R(\rho) = C_1 J_n(\sqrt{\lambda} \rho) + C_2 N_n(\sqrt{\lambda} \rho)$

$\because |R(\rho)| < +\infty \therefore C_2 = 0$

$R'(\rho_0) = C_1 \sqrt{\lambda} J_n'(\sqrt{\lambda} \rho_0) = 0$  设  $\alpha_m$  是  $J_n'(x)$  的正零点,  $m = 1, 2, 3, \dots$

$\therefore \sqrt{\lambda} \rho_0 = \alpha_m \therefore \lambda = \left(\frac{\alpha_m}{\rho_0}\right)^2, R(\rho) = C_1 J_n\left(\frac{\rho \alpha_m}{\rho_0}\right), m = 1, 2, 3, \dots$

(3) 
$$\begin{cases} (\rho R_m')' + (\lambda_m \rho - \frac{n^2}{\rho}) R_m = 0 \\ (\rho R_k')' + (\lambda_k \rho - \frac{n^2}{\rho}) R_k = 0 \end{cases} \Rightarrow R_k (\rho R_m')' - R_m (\rho R_k')' + (\lambda_m - \lambda_k) \rho R_m R_k = 0$$

即  $(\lambda_m - \lambda_k) \rho R_m R_k = R_m (\rho R_k')' - R_k (\rho R_m')'$

$$\therefore \int_0^{\rho_0} (\lambda_m - \lambda_k) \rho R_m R_k d\rho = \int_0^{\rho_0} R_m (\rho R_k')' d\rho - \int_0^{\rho_0} R_k (\rho R_m')' d\rho$$
  

$$= (\rho R_m R_k') \Big|_0^{\rho_0} - \int_0^{\rho_0} \rho R_m' R_k' d\rho + \int_0^{\rho_0} \rho R_m' R_k' d\rho - (\rho R_k R_m') \Big|_0^{\rho_0}$$

13.  $= 0$

(1) 令  $t = 2x, x^2 y'' + x y' + (4x^2 - 2)y = 0 \Rightarrow t^2 y''(t) + t y'(t) + (t^2 - 2)y = 0$

$r = \sqrt{2}$ , 不为整数  $\therefore y(t) = C_1 J_{\sqrt{2}}(t) + C_2 J_{-\sqrt{2}}(t) \Rightarrow y(x) = C_1 J_{\sqrt{2}}(2x) + C_2 J_{-\sqrt{2}}(2x)$

(2) 令  $y(x) = x u(x), x^2 y'' - x y' + (x^2 - n^2)y = 0 \Rightarrow x^2 u'' + x u' + (x^2 - n^2)u = 0$

$r = n$ , 为整数  $\therefore u(x) = C_1 J_n(x) + C_2 N_n(x) \Rightarrow y(x) = C_1 x J_n(x) + C_2 x N_n(x)$

(3) 令  $x = it, x^2 y'' + x y' - (x^2 + n^2)y = 0 \Rightarrow t^2 y''(t) + t y'(t) + (t^2 - n^2)y(t) = 0$

$r = n$ , 为整数  $\therefore y(t) = C_1 J_n(t) + C_2 N_n(t) \Rightarrow y(x) = C_1 J_n(-ix) + C_2 N_n(-ix)$

(4) 令  $y(x) = x^{-\frac{1}{2}} u(x), x^2 y'' + 2x y' + [x^2 - r(r+1)]y = 0 \Rightarrow x^2 u'' + x u' + [x^2 - (r+\frac{1}{2})^2]u = 0$

$r' = r + \frac{1}{2} \therefore u(x) = \begin{cases} C_1 J_{r+\frac{1}{2}}(x) + C_2 J_{-(r+\frac{1}{2})}(x), & r+\frac{1}{2} \notin \mathbb{N} \\ C_1 J_{r+\frac{1}{2}}(x) + C_2 N_{r+\frac{1}{2}}(x), & r+\frac{1}{2} \in \mathbb{N} \end{cases}$

$\Rightarrow y(x) = \begin{cases} \frac{1}{\sqrt{x}} [C_1 J_{r+\frac{1}{2}}(x) + C_2 J_{-(r+\frac{1}{2})}(x)], & r+\frac{1}{2} \notin \mathbb{N} \\ \frac{1}{\sqrt{x}} [C_1 J_{r+\frac{1}{2}}(x) + C_2 N_{r+\frac{1}{2}}(x)], & r+\frac{1}{2} \in \mathbb{N} \end{cases}$

14.

(1) 
$$\int (x^3 + x) J_0 dx = \int x^2 [x J_0(x)] dx + \int x J_0(x) dx$$
  

$$= x^3 J_1(x) - \int 2x^2 J_1(x) dx + \int x J_0(x) dx$$

$$= x^3 J_1(x) - 2x^2 J_2(x) + x J_1(x) + C$$

$$= (x^3 + x) J_1(x) - 2x^2 J_2(x) + C = (x^3 - 3x) J_1(x) + 2x^2 J_0(x) + C$$

(2) 
$$\int (1-x^2) J_1(x) dx = \int \frac{J_1(x)}{x^2} dx - \int x^2 J_1(x) dx$$

$$= -J_0(x) - x^2 J_2(x) + C$$

15. 
$$A_m = \frac{2}{[J_0'(\mu_m^{(2)})]^2} \int_0^1 \rho f(\rho) J_0(\mu_m^{(2)} \rho) d\rho$$

$$= \frac{2}{[J_0'(\mu_m^{(2)})]^2} \int_0^1 \rho^3 J_0(\mu_m^{(2)} \rho) d\rho$$

$$= \frac{2}{J_1^2(\mu_m^{(2)})} \times \frac{(\mu_m^{(2)3} J_1(\mu_m^{(2)}) - 2\mu_m^{(2)2} J_2(\mu_m^{(2)}))}{\mu_m^{(2)2}}$$

$$= \frac{2}{J_1^2(\mu_m^{(2)})} \frac{(\mu_m^{(2)3} - 4\mu_m^{(2)}) J_1(\mu_m^{(2)})}{\mu_m^{(2)2}}$$

$$= \frac{2[\mu_m^{(2)2} - 4]}{(\mu_m^{(2)})^3 J_1(\mu_m^{(2)})}, m \geq 1$$

$$\therefore f(\rho) = \sum_{m=1}^{\infty} A_m J_m(\mu_m^{(2)} \rho)$$

$$\begin{aligned}
 16. A_m &= \frac{2}{(1 - \frac{4}{\alpha_m^2}) J_2'(\alpha_m)} \int_0^1 x f(x) J_2(\alpha_m x) dx \\
 &= \frac{2}{(1 - \frac{4}{\alpha_m^2}) J_2'(\alpha_m)} \int_0^1 x^3 J_2(\alpha_m x) dx \\
 &= \frac{2}{(1 - \frac{4}{\alpha_m^2}) J_2'(\alpha_m)} \frac{\alpha_m^3 J_3(\alpha_m)}{\alpha_m^4} \\
 &= \frac{2\alpha_m J_3(\alpha_m)}{(\alpha_m^2 - 4) J_2'(\alpha_m)}, m \geq 1
 \end{aligned}$$

$$\therefore f(x) = \sum_{m=1}^{\infty} A_m J_2(\alpha_m x)$$

$$17. \text{平方模: } \int_0^2 x J_2^2(\alpha_m x) dx = 2 J_1^2(2\alpha_m)$$

$$\begin{aligned}
 A_m &= \frac{1}{2 J_1^2(2\alpha_m)} \int_0^2 x f(x) J_2(\alpha_m x) dx \\
 &= \frac{1}{2 J_1^2(2\alpha_m)} \int_0^1 x J_2(\alpha_m x) dx \\
 &= \frac{1}{2 J_1^2(2\alpha_m)} \frac{\alpha_m J_1(\alpha_m)}{\alpha_m^2} \\
 &= \frac{J_1(\alpha_m)}{2\alpha_m J_1^2(2\alpha_m)}, m \geq 1
 \end{aligned}$$

$$\therefore f(x) = \sum_{m=1}^{\infty} A_m J_2(\alpha_m x)$$

$$18. \text{时空分离: } W(x, y) T'(t) - a^2 W''(x, y) \cdot T(t) = 0 \Rightarrow \frac{T'(t)}{a^2 T(t)} = \frac{W''(x, y)}{W(x, y)} = -\mu$$

$$\text{空间分离: 设 } X''(x) + \lambda X(x) = 0, \text{ 则 } Y''(y) + (\mu - \lambda) Y(y) = 0$$

$$\text{对于 } X(x): X(0) = X(1) = 0 \quad \therefore \lambda_n = (n\pi)^2, X_n(x) = \sin n\pi x, n = 1, 2, 3, \dots$$

$$\text{对于 } Y(y): Y(0) = Y(2) = 0 \quad \therefore \mu - \lambda_n = (\frac{m\pi}{2})^2, Y_m(y) = \sin \frac{m\pi y}{2}, m = 1, 2, 3, \dots$$

$$\therefore \mu_{m,n} = (n\pi)^2 + (\frac{m\pi}{2})^2, W(x, y)_{n,m} = (\sin n\pi x) (\sin \frac{m\pi}{2} y)$$

$$T'(t) + a^2 \mu T(t) = 0 \Rightarrow T'_{m,n}(t) + a^2 \mu_{m,n} T_{m,n}(t) = 0$$

$$\text{设 } T_{m,n} = C_{m,n} \exp(-a^2 \mu_{m,n} t)$$

$$\varphi_{m,n} = \frac{2 \times 2}{1 \times 2} \int_0^1 \int_0^2 xy (\sin n\pi x) (\sin \frac{m\pi}{2} y) dx dy = \frac{8(-1)^{m+n}}{mn\pi^2}$$

$$T_{m,n}(0) = C_{m,n} = \varphi_{m,n}$$

$$\therefore u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (\sin n\pi x) (\sin \frac{m\pi}{2} y) \varphi_{m,n} \exp(-a^2 \mu_{m,n} t)$$

$$\text{其中: } \varphi_{m,n} = \frac{8(-1)^{m+n}}{mn\pi^2}, \mu_{m,n} = (n\pi)^2 + (\frac{m\pi}{2})^2$$

$$19. \text{以柱坐标系建立定解问题: } \begin{cases} a^2(u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + u_{zz}) = 0 \\ u(\rho, 0) = u(1, z) = 0 \\ u(\rho, 2) = 3 \end{cases}$$

$$\text{设 } u(\rho, z) = R(\rho) Z(z)$$

$$\therefore (R''(\rho) + \frac{1}{\rho} R'(\rho)) Z(z) + R(\rho) + Z''(z) \quad \text{设 } \frac{R''(\rho) + \frac{1}{\rho} R'(\rho)}{R(\rho)} = -\frac{Z''(z)}{Z(z)} = -\lambda$$

$$R''(\rho) + \frac{1}{\rho} R'(\rho) + \lambda R(\rho) = 0$$

$$\text{即 } \begin{cases} \rho^2 R''(\rho) + R(\rho) + \lambda \rho^2 R(\rho) = 0 \\ R(1) = 0 \end{cases} \therefore \lambda_m = (\mu_m^{(1)})^2, R_m(\rho) = J_0(\mu_m^{(1)} \rho), m=1, 2, 3, \dots$$

$$Z_m''(z) - \lambda_m Z_m(z) = 0 \Rightarrow Z_m(z) = C_{1m} \cosh \mu_m^{(1)} z + C_{2m} \sinh \mu_m^{(1)} z$$

$$\begin{cases} Z_m(0) = C_{1m} = 0 \end{cases}$$

$$\begin{cases} Z_m(2) = C_{2m} \sinh 2\mu_m^{(1)} = \frac{2}{[J_1(\mu_m^{(1)})]^2} \int_0^1 3\rho J_0(\mu_m^{(1)} \rho) d\rho = \frac{2}{[J_1(\mu_m^{(1)})]^2} \frac{3}{[\mu_m^{(1)}]^2} \mu_m^{(1)} J_1(\mu_m^{(1)}) \\ = \frac{6}{\mu_m^{(1)} J_1(\mu_m^{(1)})} \end{cases}$$

$$\therefore C_{1m} = 0$$

$$\begin{cases} C_{2m} = \frac{6}{\mu_m^{(1)} J_1(\mu_m^{(1)}) \sinh 2\mu_m^{(1)}} \end{cases}$$

$$\therefore u(\rho, z) = \sum_{m=1}^{\infty} \frac{6 \sinh \mu_m^{(1)} z J_0(\mu_m^{(1)} \rho)}{\mu_m^{(1)} J_1(\mu_m^{(1)}) \sinh 2\mu_m^{(1)}}$$

20. 设  $u(\rho, t) = T(t)R(\rho)$

$$R(\rho) \text{ 对应的齐次方程: } \begin{cases} R''(\rho) + \frac{1}{\rho} R'(\rho) + \lambda R(\rho) = 0 \\ R(\rho_0) = 0, |R(\rho)| < +\infty \end{cases} \therefore \begin{cases} \lambda_m = (\frac{\mu_m^{(1)}}{\rho_0})^2 \\ R_m(\rho) = J_0(\frac{\mu_m^{(1)}}{\rho_0} \rho) \end{cases}, m=1, 2, 3, \dots$$

$$B_m = \frac{2}{\rho_0^2 [J_1(\frac{\mu_m^{(1)}}{\rho_0})]^2} \int_0^{\rho_0} B \rho J_0(\frac{\mu_m^{(1)}}{\rho_0} \rho) d\rho = \frac{2B}{\rho_0^2 [J_1(\frac{\mu_m^{(1)}}{\rho_0})]^2} \times [\frac{\rho_0}{\mu_m^{(1)}}]^2 \mu_m^{(1)} J_1(\mu_m^{(1)}) = \frac{2B}{\mu_m^{(1)} J_1(\mu_m^{(1)})}$$

$$T_m''(t) + a^2 \lambda_m T_m(t) = B_m \quad \text{设 } T_m(t) = C_{1m} \cos(a \frac{\mu_m^{(1)}}{\rho_0} t) + C_{2m} \sin(a \frac{\mu_m^{(1)}}{\rho_0} t) + \frac{B_m \rho_0^2}{a^2 \mu_m^{(1)2}}$$

$$\begin{cases} T_m(0) = C_{1m} + \frac{B_m \rho_0^2}{a^2 \mu_m^{(1)2}} = 0 \end{cases}$$

$$\begin{cases} T_m'(0) = a \frac{\mu_m^{(1)}}{\rho_0} C_{2m} = 0 \end{cases}$$

$$\therefore \begin{cases} C_{1m} = -\frac{B_m \rho_0^2}{a^2 \mu_m^{(1)2}} \\ C_{2m} = 0 \end{cases}$$

$$\therefore T_m(t) = \frac{B_m \rho_0^2}{a^2 \mu_m^{(1)2}} [1 - \cos(a \frac{\mu_m^{(1)}}{\rho_0} t)]$$

$$\therefore u(\rho, z) = \sum_{m=1}^{\infty} \frac{B_m \rho_0^2}{a^2 \mu_m^{(1)2}} [1 - \cos(a \frac{\mu_m^{(1)}}{\rho_0} t)] J_0(\frac{\mu_m^{(1)}}{\rho_0} \rho), \text{ 其中 } B_m = \frac{2B}{\mu_m^{(1)} J_1(\mu_m^{(1)})}$$

补充题1 (例 3.11)

$$\text{柱坐标下: } \begin{cases} u_t = a^2 (u_{\rho\rho} + \frac{1}{\rho} u_{\rho}) , 0 < \rho < 1, t > 0 \\ u(1, t) = 0, t \geq 0 \\ u(\rho, 0) = \varphi(\rho), 0 \leq \rho \leq 1 \end{cases}$$

$$\text{设 } u(\rho, t) = R(\rho)T(t) \therefore \frac{R'' + \frac{R'}{\rho}}{R} = \frac{T'}{a^2 T} = -\lambda$$

$$\Rightarrow \begin{cases} R''(\rho) + \frac{R'}{\rho} + \lambda R = 0 \\ R(1) = 0 \end{cases} \Rightarrow \begin{cases} \lambda_m = (\mu_m^{(1)})^2 \\ R_m(\rho) = J_0(\mu_m^{(1)} \rho) \end{cases}, m=1, 2, 3, \dots$$

$$T_m'(t) + a^2 \lambda_m T_m(t) = 0 \therefore T_m = A_m \exp(-a^2 (\mu_m^{(1)})^2 t)$$

$$T_m(0) = A_m = \frac{1}{[J_1(\mu_m^{(1)})]^2} \int_0^1 \rho \varphi(\rho) J_0(\mu_m^{(1)} \rho) d\rho$$

$$\therefore u(\rho, t) = \sum_{m=1}^{\infty} A_m e^{-a^2 (\mu_m^{(1)})^2 t} J_0(\mu_m^{(1)} \rho)$$

补充题2 (例 3.13)

$$\text{定解问题 } \begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + u_{zz} = 0, 0 < \rho < a, 0 < z < h \\ u(\rho, 0) = 0, u(\rho, h) = \rho^2, \rho \leq a \\ u(a, z) = 0, 0 \leq z \leq h \end{cases}$$

$$\text{令 } u(\rho, z) = R(\rho)Z(z) \text{ 设 } \frac{R'' + \frac{1}{\rho} R'}{R} = -\frac{Z''(z)}{Z(z)} = -\lambda$$

$$\therefore \begin{cases} R''(\rho) + \frac{1}{\rho} R'(\rho) + \lambda R = 0 \\ R(a) = 0, |R(\rho)| < +\infty \end{cases} \therefore \begin{cases} \lambda_m = (\frac{\mu_m^{(1)}}{a})^2 \\ R(\rho) = J_0(\frac{\mu_m^{(1)}}{a} \rho) \end{cases}, m=1, 2, 3, \dots$$

$$Z_m''(z) - \lambda_m Z_m(z) = 0$$

$$\therefore \text{设 } Z_m(z) = a_m \cosh \frac{\mu_m^{(2)}}{a} z + b_m \sinh \frac{\mu_m^{(2)}}{a} z$$

$$\begin{cases} Z_m(0) = a_m = 0 \\ Z_m(h) = b_m \sinh \frac{\mu_m^{(2)}}{a} h = \frac{2}{a^2 [J_0'(\mu_m^{(2)})]^2} \int_0^a \rho^3 J_0\left(\frac{\mu_m^{(2)}}{a} \rho\right) d\rho = \frac{2}{a^2 [J_1(\mu_m^{(2)})]^2} \times \frac{a^4}{(\mu_m^{(2)})^4} \times [(\mu_m^{(2)})^3 J_1(\mu_m^{(2)}) - 4 \mu_m^{(2)} J_1(\mu_m^{(2)})] = \frac{2a^2 [(\mu_m^{(2)})^2 - 4]}{(\mu_m^{(2)})^3 J_1(\mu_m^{(2)})} \end{cases}$$

$$\therefore \begin{cases} a_m = 0 \\ b_m = \frac{2a^2 [(\mu_m^{(2)})^2 - 4]}{(\mu_m^{(2)})^3 J_1(\mu_m^{(2)}) \sinh \frac{\mu_m^{(2)}}{a} h} \end{cases}$$

$$\therefore Z_m(z) = b_m \sinh \frac{\mu_m^{(2)}}{a} z$$

$$\therefore u(\rho, z) = \sum_{m=1}^{\infty} b_m \sinh \frac{\mu_m^{(2)}}{a} z J_0\left(\frac{\mu_m^{(2)}}{a} \rho\right), \text{ 其中 } b_m = \frac{2a^2 [(\mu_m^{(2)})^2 - 4]}{(\mu_m^{(2)})^3 J_1(\mu_m^{(2)}) \sinh \frac{\mu_m^{(2)}}{a} h}$$

# 习题6

1.

(1) 特征线方程:  $\frac{dx}{dt} - 2 = 0 \Rightarrow x = 2t + \tau$

将原定解问题化简为  $\begin{cases} \frac{du}{dt} = 0 \\ u(0) = \tau^2 \end{cases} \Rightarrow u(x, t) = u(\tau) = \tau^2 = (x - 2t)^2$

(2) 特征向量场:  $\vec{\alpha} = (2, 1, xt - u)$

将原定解问题化简为  $\begin{cases} \frac{dx}{ds} = 2, x(0) = \tau \\ \frac{dt}{ds} = 1, t(0) = 0 \\ \frac{du}{ds} = xt - u, u(0) = 2 - \tau \end{cases} \Rightarrow \begin{cases} x = 2s + \tau \\ t = s \\ u = -2e^{-s} + 2s^2 + (\tau - 4)(s - 1) \end{cases}$  消去  $s, \tau$ :  $u = -2e^{-t} + xt - x - 2t + 4$

(3) 特征向量场:  $\vec{\alpha} = (1, -2, xu)$

将原定解问题化简为  $\begin{cases} \frac{dx}{ds} = 1, x(0) = \tau \\ \frac{dt}{ds} = -2, t(0) = 0 \\ \frac{du}{ds} = xu, u(0) = 2\tau e^{\frac{\tau}{2}} \end{cases} \Rightarrow \begin{cases} x = s + \tau \\ t = -2s \\ u = 2\tau e^{\frac{t}{2}} \end{cases}$  消去  $s, \tau$ :  $u = (2x + t)e^{\frac{x^2}{2}}$

(4) 特征向量场:  $\vec{\alpha} = (1 + x^2, 1, u)$

将原定解问题化简为  $\begin{cases} \frac{dx}{ds} = 1 + x^2, x(0) = \tau \\ \frac{dt}{ds} = 1, t(0) = 0 \\ \frac{du}{ds} = u, u(0) = \arctan \tau \end{cases} \Rightarrow \begin{cases} \arctan x = s + \arctan \tau \\ t = s \\ u = \arctan \tau e^s \end{cases}$  消去  $s, \tau$ :  $u = (\arctan x - t)e^t$

2. 特征方程  $(\frac{dx}{dt})' - 1 = 0 \Rightarrow$  特征线:  $x + t = \tau_1, x - t = \tau_2$

变量代换:  $\xi = x + t, \eta = x - t \therefore u_{tt} - u_{xx} = -4u_{\xi\eta} = 0$

$\therefore u(\xi, \eta) = f(\xi) + g(\eta)$  即  $u(x, t) = f(x + t) + g(x - t)$

$\begin{cases} u|_{x=0} = f(t) + g(-t) = \varphi(t) \\ u|_{x=t} = f(2t) + g(0) = \psi(t) \end{cases} \Rightarrow \begin{cases} f(t) = \frac{1}{2}(\varphi(\frac{t}{2}) + g(0)) \\ g(t) = \varphi(-t) - \varphi(-\frac{t}{2}) + g(0) \end{cases}$

$\therefore u(x, t) = \varphi(\frac{t-x}{2}) + \psi(\frac{x+t}{2}) - \varphi(\frac{t-x}{2})$

3. 特征方程  $(\frac{dx}{dt})' - 1 = 0 \Rightarrow$  特征线:  $x + t = \tau_1, x - t = \tau_2$

变量代换:  $\xi = x + t, \eta = x - t \therefore u_{tt} - u_{xx} = -4u_{\xi\eta} = 0$

$u(\xi, \eta) = f(\xi) + g(\eta)$  即  $u(x, t) = f(x + t) + g(x - t)$

$\begin{cases} u|_{x=-t} = f(0) + g(-2t) = \varphi(t) \\ u|_{x=t} = f(2t) + g(0) = \psi(t) \end{cases} \Rightarrow \begin{cases} f(t) = \frac{1}{2}(\varphi(\frac{t}{2}) + g(0)) \\ g(t) = \varphi(-\frac{t}{2}) - f(0) \end{cases}$

$\therefore u(x, t) = \varphi(\frac{t-x}{2}) + \psi(\frac{x+t}{2}) - (f(0) + g(0)) = \varphi(\frac{t-x}{2}) + \psi(\frac{x+t}{2}) - \varphi(0)$

4.  $\begin{cases} u(x, t) = f(x+t) + g(x-t) \\ \varphi(x) = 0 \\ \psi(x) = \sin x \end{cases} \therefore u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} \sin \alpha d\alpha = \sin x \sin t$

5.  $u_x = u_\xi - 3u_\eta, u_y = u_\xi + u_\eta$

$u_{xx} = u_{\xi\xi} - 6u_{\xi\eta} + 9u_{\eta\eta}, u_{xy} = u_{\xi\xi} - 2u_{\xi\eta} - 3u_{\eta\eta}, u_{yy} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$

$\therefore u_{xx} + 2u_{xy} - 3u_{yy} = -16u_{\xi\eta} = 0$

$\therefore u(x, y) = f(x+y) + g(y-3x)$

$u(x, 0) = f(x) + g(-3x) = \varphi(x)$

$u_y(x, 0) = f'(x) + g'(-3x) = \psi(x)$

对  $[0, x]$  区间积分:  $\int_0^x \psi(\alpha) d\alpha = \int_0^x f'(\alpha) d\alpha - \frac{1}{3} \int_0^{-3x} g'(\alpha) d\alpha = f(x) - \frac{1}{3}g(-3x) - [f(0) - \frac{1}{3}g(0)]$

设  $C_0 = [f(0) - \frac{1}{3}g(0)]$

$\therefore \begin{cases} f(x) + g(-3x) = \varphi(x) \\ f(x) - \frac{1}{3}g(-3x) = \int_0^x \psi(\alpha) d\alpha + C_0 \end{cases} \Rightarrow \begin{cases} f(x) = \frac{1}{4}\varphi(x) + \frac{3}{4} \int_0^x \psi(\alpha) d\alpha + \frac{3}{4}C_0 \\ g(x) = \frac{3}{4}\varphi(-\frac{x}{3}) - \frac{1}{4} \int_0^{-\frac{x}{3}} \psi(\alpha) d\alpha - \frac{3}{4}C_0 \end{cases}$

$u(x, y) = f(x+y) + g(y-3x)$

$= \frac{1}{4}\varphi(x+y) + \frac{3}{4}(x - \frac{y}{3}) + \frac{3}{4} \int_{x-\frac{y}{3}}^{x+y} \psi(\alpha) d\alpha$

6. 变量代换:  $\xi = x+y, \eta = y-3x$

$$u_{xx} + 2u_{xy} - 3u_{yy} = -16u_{xy} = \frac{5}{3} \therefore u_{xy} = -\frac{1}{16} \frac{5}{3}$$

$$\therefore u_{xy} = -\frac{1}{16} \frac{5}{3} + f(\frac{5}{3}) \therefore u = -\frac{1}{32} \frac{5}{3} y + F(\frac{5}{3}) + G(y)$$

$$\therefore u(x, y) = -\frac{1}{32} (x+y)(y-3x) + F(x+y) + G(y-3x)$$

$$\begin{cases} u(x, 0) = F(x) + G(-3x) + \frac{3}{32} x^2 = 0 \\ u_y(x, 0) = F'(x) + G'(-3x) + \frac{5}{32} x^2 = 0 \end{cases}$$

$$\text{在 } [0, x] \text{ 区间积分: } \int_0^x \frac{5}{32} \alpha^2 d\alpha = \int_0^x F'(\alpha) d\alpha - \frac{1}{3} \int_0^{-3x} G'(\alpha) d\alpha$$

$$\therefore \begin{cases} F(x) + G(-3x) = -\frac{3}{32} x^3 \\ F(x) - \frac{1}{3} G(-3x) = -\frac{5}{96} x^3 + C_0 \end{cases} \text{ 其中, } C_0 = F(0) - \frac{1}{3} G(0)$$

$$\therefore \begin{cases} F(x) = -\frac{1}{16} x^3 + \frac{3}{4} C_0 \\ G(x) = \frac{1}{32 \times 2} x^3 - \frac{3}{4} C_0 \end{cases} \therefore u(x, y) = -\frac{1}{32} (x+y)^3 (y-3x) - \frac{1}{16} (x+y)^3 + \frac{1}{32 \times 2} (y-3x)^3 = -\frac{1}{6} x y^2 - \frac{5}{54} y^3$$

$$\therefore \begin{cases} F(x) = -\frac{1}{16} x^3 + \frac{3}{4} C_0 \\ G(x) = \frac{1}{32 \times 2} x^3 - \frac{3}{4} C_0 \end{cases} \therefore u(x, y) = -\frac{1}{32} (x+y)^3 (y-3x) - \frac{1}{16} (x+y)^3 + \frac{1}{32 \times 2} (y-3x)^3 = -\frac{1}{6} x y^2 - \frac{5}{54} y^3$$

$$7. u(x, y) = \frac{1}{6} x^2 y^2 + f(x) + g(y)$$

$$\begin{cases} u(x, 0) = f(x) + g(0) = x^2 \\ u(1, y) = \frac{1}{6} y^2 + f(1) + g(y) = 0.5y \end{cases} \Rightarrow \begin{cases} f(x) = x^2 \\ g(y) = 0.5y - \frac{1}{6} y^2 - 1 \end{cases}$$

$$\therefore u(x, y) = \frac{1}{6} x^2 y^2 + x^2 - \frac{1}{6} y^2 + 0.5y - 1$$

8. 问题(2)提法错误

$$\text{特征线斜率 } \frac{dt}{dx} = -\frac{1}{a} < 0 \therefore u(x, t) \text{ 由 } u(x, 0) \text{ 决定}$$

又给定  $u(1, t)$ , 产生矛盾,

$\therefore$  问题(2)提法错误

$$9. \text{特征向量场: } \vec{\alpha} = (-1, 1-x-u, 1)$$

$$\text{将原定解问题化简为 } \begin{cases} \frac{dx}{ds} = -1, x(0) = \tau \\ \frac{dt}{ds} = 1-x-u, t(0) = 0 \\ \frac{du}{ds} = 1, u(0) = 0 \end{cases} \Rightarrow \begin{cases} x = -s + \tau \\ t = (1-\tau)s \\ u = s \end{cases} \text{ 消去 } s, \tau: u = \begin{cases} \frac{1}{2} [(1-x) - \sqrt{(1-x)^2 - 4t}], x < 1, 4t \leq (1-x)^2 \\ \frac{1}{2} [(1-x) + \sqrt{(1-x)^2 - 4t}], x > 1, 4t \leq (1-x)^2 \end{cases}$$

$$10. \text{特征方程组 } \begin{cases} \frac{dx}{ds} = u, x(0) = \tau \\ \frac{dt}{ds} = 1, t(0) = 0 \\ \frac{du}{ds} = 0, u(0) = \varphi(\tau) \end{cases} \text{ 其中 } \varphi(\tau) = \begin{cases} 1, \tau < 0 \\ 1-\tau, 0 \leq \tau \leq 1 \\ 0, \tau > 1 \end{cases} \therefore \begin{cases} x = \tau + \varphi(\tau) \cdot s \\ t = s \\ u = \varphi(\tau) \end{cases}$$

$$\text{消去 } s, \tau \text{ 得: } u(x, t) = \begin{cases} 1, x < t \\ \frac{1-x}{1-t}, t \leq x \leq 1 \\ 0, x > 1 \end{cases}$$

$$11. u_x + \frac{3}{2} u_y + \frac{5}{2} u_z - \frac{1}{2} u = 0$$

$$\therefore \text{特征方程 } \begin{cases} \frac{dy}{dx} = \frac{3}{2} \\ \frac{dz}{dx} = \frac{5}{2} \end{cases} \Rightarrow \text{特征线 } \begin{cases} y = \frac{3}{2} x + \tau_1 \\ z = \frac{5}{2} x + \tau_2 \end{cases}$$

$$\therefore \text{化简原定解问题 } \begin{cases} \frac{du}{dx} - \frac{1}{2} u = 0, x > 0 \\ u(0, y(0), z(0)) = \varphi(\tau_1, \tau_2) \end{cases} \Rightarrow u(x, y(x), z(x)) = \varphi(\tau_1, \tau_2) e^{\frac{x}{2}}$$

$$\therefore u(x, y, z) = \varphi(y - \frac{3}{2} x, z - \frac{5}{2} x) e^{\frac{x}{2}}$$

$$12. \text{特征方程 } \begin{cases} \frac{dy}{dx} = z \\ \frac{dz}{dx} = 6x \end{cases} \Rightarrow \text{特征线 } \begin{cases} y = xz + \tau_1 \\ z = 3x^2 + \tau_2 \end{cases}$$

$$\text{化简原定解问题 } \begin{cases} \frac{du}{dx} = 0, x > 0 \\ u(0, y(0), z(0)) = \varphi(\tau_1, \tau_2) \end{cases} \Rightarrow u(x, y(x), z(x)) = \varphi(\tau_1, \tau_2)$$

$$\therefore u(x, y, z) = \varphi(y - xz, z - 3x^2)$$

补充题

$$1. \begin{cases} u_t + u \cdot u_x = 0, t > 0, -\infty < x < \infty \\ u(x, 0) = \frac{1}{2} + \sin x, -\infty < x < \infty \end{cases}$$

$$\begin{cases} \frac{dx}{ds} = u, x(0) = \tau \\ \frac{dt}{ds} = 1, t(0) = 0 \\ \frac{du}{ds} = 0, u(0) = \frac{1}{2} + \sin \tau \end{cases} \Rightarrow \begin{cases} x = u s + \tau \\ t = s \\ u = \frac{1}{2} + \sin \tau \end{cases} \text{ 消去 } s, \tau: u(x, t) = \frac{1}{2} + \sin[x - t \cdot u(x, t)]$$

## 2. D'Alembert 公式的运用

$$\textcircled{1} \begin{cases} u_{tt} - a^2 u_{xx} = 0, t > 0, -\infty < x < +\infty \\ u(x, 0) = \sin x, u_t(x, 0) = ac \cdot \sin x, -\infty < x < +\infty \end{cases}$$

$$\begin{aligned} u(x, t) &= \frac{1}{2} [\phi(x-at) + \phi(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha \\ &= \frac{1}{2} [\sin(x-at) + \sin(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} ac \cdot \sin \alpha d\alpha \\ &= \frac{1}{2} [\sin(x-at) + \sin(x+at)] + \frac{1}{2} [\sin(x+at) - \sin(x-at)] \\ &= \sin(x+at) \end{aligned}$$

$$\textcircled{2} \begin{cases} u_{tt} - a^2 u_{xx} = 0, t > 0, -\infty < x < +\infty \\ u(x, 0) = \sin x, u_t(x, 0) = x, -\infty < x < +\infty \end{cases}$$

$$\begin{aligned} u(x, t) &= \frac{1}{2} [\phi(x-at) + \phi(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha \\ &= \frac{1}{2} [\sin(x-at) + \sin(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \alpha d\alpha \\ &= \sin x (\cos at + xt) \end{aligned}$$

# 习题 5

1. 利用 Green 函数法求解定解问题

$$(1) \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & -\infty < x < +\infty, y > 0 \\ u = \varphi(x), & -\infty < x < +\infty, y = 0 \end{cases}$$

取  $P(x, y), P_0(\xi, \eta) \in \Omega, P_1(\xi, -\eta)$

$$\therefore G(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_0} - \frac{1}{2\pi} \ln \frac{1}{r_1}, \text{ 其中 } r_0 = \sqrt{(x-\xi)^2 + (y-\eta)^2}, r_1 = \sqrt{(x-\xi)^2 + (y+\eta)^2}$$

$$\frac{\partial G}{\partial n} = -\frac{\partial G}{\partial y} \Big|_{y=0} = -\left[ -\frac{1}{2\pi} \frac{y-\eta}{r_0^2} + \frac{1}{2\pi} \frac{y+\eta}{r_1^2} \right]_{y=0} = \frac{-\eta}{\pi[(x-\xi)^2 + \eta^2]}$$

$$\therefore u(\xi, \eta) = -\int_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iint_{\Omega} G f d\sigma = \frac{\eta}{\pi} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{(x-\xi)^2 + \eta^2} dx$$

$$(2) \begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega, \Omega = \{(x, y, z) | z > 0\} \\ u(x, y, 0) = \varphi(x, y), & (x, y) \in \partial\Omega \end{cases}$$

取  $P(x, y, z), P_0(\xi, \eta, \zeta) \in \Omega, P_1(\xi, \eta, -\zeta)$

$$G(P, P_0) = \frac{1}{4\pi r_0} - \frac{1}{4\pi r_1}, \text{ 其中 } r_0 = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}, r_1 = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2}$$

$$\frac{\partial G}{\partial n} = -\frac{\partial G}{\partial z} \Big|_{z=0} = \frac{-\zeta}{2\pi[(x-\xi)^2 + (y-\eta)^2 + \zeta^2]^{3/2}}$$

$$\therefore u(\xi, \eta, \zeta) = -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

$$= \frac{\zeta}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{dx dy}{[(x-\xi)^2 + (y-\eta)^2 + \zeta^2]^{3/2}} + \int_0^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G(P, P_0) f(x, y, z) dx dy dz$$

2. 利用镜像法求出下述区域的 Green 函数

$$(1) \text{ 第一象限 } \Omega = \{(x, y) | x \geq 0, y \geq 0\}, \partial\Omega = \{(x, y) | x=0, y \geq 0\} \cup \{(x, y) | y=0, x \geq 0\}$$

取  $P(x, y), P_0(\xi, \eta) \in \Omega, P_1(\xi, -\eta), P_2(-\xi, \eta), P_3(-\xi, -\eta)$

$$G(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_0} - \frac{1}{2\pi} \ln \frac{1}{r_1} - \frac{1}{2\pi} \ln \frac{1}{r_2} + \frac{1}{2\pi} \ln \frac{1}{r_3} = \frac{1}{2\pi} \ln \frac{\sqrt{[(x-\xi)^2 + (y-\eta)^2]} \sqrt{[(x+\xi)^2 + (y+\eta)^2]}}{\sqrt{[(x-\xi)^2 + (y+\eta)^2]} \sqrt{[(x+\xi)^2 + (y-\eta)^2]}}$$

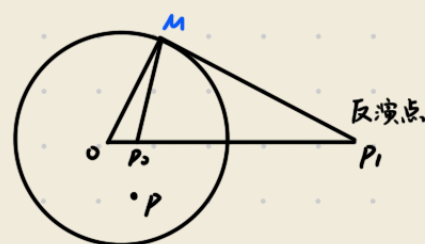
$$(2) \text{ 证明圆域上的 Green 函数可表示为 } G(P, P_0) = \frac{1}{2\pi} \left[ \ln \left( \frac{1}{r_{P_0 P}} \right) - \ln \left( \frac{R}{|OP_0|} \frac{1}{r_{P_1 P}} \right) \right]$$

$$\because |OP_0| \cdot |OP_1| = R^2 \therefore \triangle OMP_0 \sim \triangle P_1M$$

$$\therefore \frac{r_{P_0 M}}{r_{P_1 M}} = \frac{|OP_0|}{|OM|} = \frac{|OP_0|}{R} \text{ 即 } \frac{1}{r_{P_0 M}} = \frac{R}{|OP_0|} \frac{1}{r_{P_1 M}}$$

$$\therefore \frac{1}{2\pi} \ln \left( \frac{1}{r_{P_0 M}} \right) - \frac{1}{2\pi} \ln \left( \frac{R}{|OP_0|} \frac{1}{r_{P_1 M}} \right) = 0$$

$$\therefore G(P, P_0) = \frac{1}{2\pi} \left[ \ln \left( \frac{1}{r_{P_0 P}} \right) - \ln \left( \frac{R}{|OP_0|} \frac{1}{r_{P_1 P}} \right) \right]$$



$$(3) \text{ 半空间 } \Omega = \{(x, y, z) | z > x + y\}$$

取  $P(x, y, z), P_0(\xi, \eta, \zeta) \in \Omega$

$$\vec{e}_n = \frac{1}{\sqrt{3}}(1, 1, -1)$$

$$P_1(\xi', \eta', \zeta'), \text{ 其中 } \xi' = \xi - 2 \frac{\xi + \eta - \zeta}{1+1+1} = \frac{1}{3}(\xi - 2\eta + 2\zeta)$$

$$\eta' = \eta - 2 \frac{\xi + \eta - \zeta}{1+1+1} = \frac{1}{3}(-2\xi + \eta + 2\zeta)$$

$$\zeta' = \zeta - 2 \frac{\xi + \eta - \zeta}{1+1+1} = \frac{1}{3}(-2\xi - 2\eta + 5\zeta)$$

$$G(P, P_0) = \frac{1}{4\pi} \left( \frac{1}{r_{PP_0}} - \frac{1}{r_{PP_1}} \right)$$

3. 设  $\Omega \subset \mathbb{R}^3$  为有界区域,  $\partial\Omega$  充分光滑, 证明下述定解问题可解的必要条件是  $\iiint_{\Omega} f dV = \iint_{\partial\Omega} \varphi ds$

$$\begin{cases} \Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ \frac{\partial u(x, y, z)}{\partial n} \Big|_{\partial\Omega} = \varphi(x, y, z), & (x, y, z) \in \partial\Omega \end{cases}$$

$$\iiint_{\Omega} \Delta u dV = \iiint_{\Omega} f(x, y, z) dV$$

$$\text{又 } \iiint_{\Omega} \Delta u dV = \iiint_{\Omega} \nabla \cdot (\nabla u) dV = \iint_{\partial\Omega} \frac{\partial u}{\partial n} ds = \iint_{\partial\Omega} \varphi ds$$

$$\therefore \iiint_{\Omega} f dV = \iint_{\partial\Omega} \varphi ds$$